

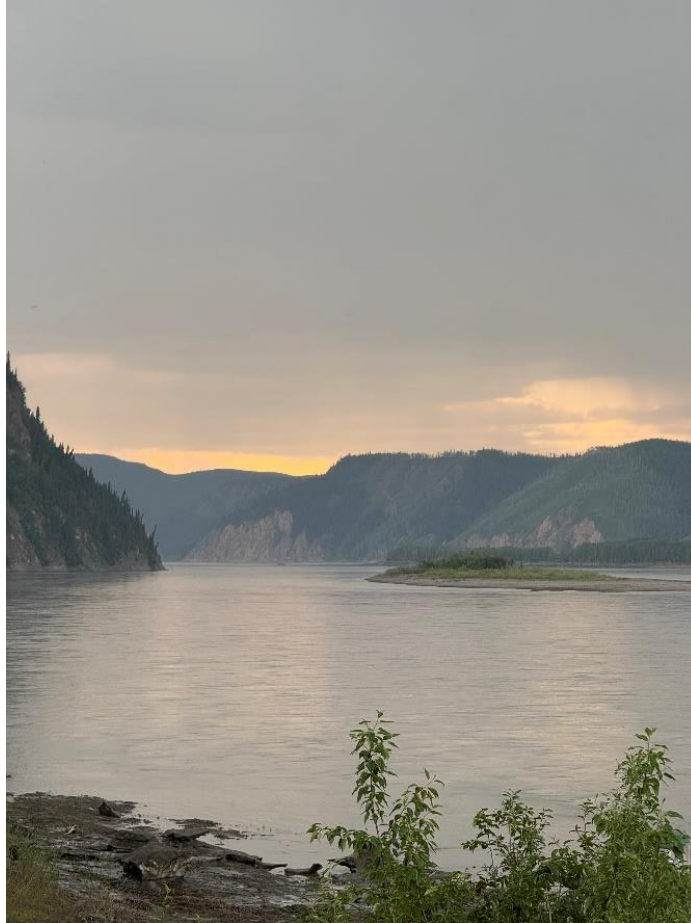
Alaska Cooperative Fish and Wildlife Research Unit

Annual Research Report: 2025



Alaska Cooperative Fish and Wildlife Research Unit
P.O. Box 757020, University of Alaska Fairbanks
Fairbanks, AK 99775-7020
uaf-iab-akcfwru@alaska.edu
<https://uaf.edu/akcfwru>





Sunset on the Yukon River near Coal Creek.
Photo credit: Thomas House.

• • •

Not for Publication: Because this report is one of progress, the data presented are often incomplete, and the conclusions reached may not be final. Consequently, permission to publish any of the information herein is withheld pending approval from the Alaska Cooperative Fish and Wildlife Research Unit.

• • •

The University of Alaska is an affirmative action/equal opportunity employer and educational institution. The University of Alaska does not discriminate on the basis of race, religion, color, national origin, citizenship, age, sex, physical or mental disability, status as a protected veteran, marital status, changes in marital status, pregnancy, childbirth or related medical conditions, parenthood, sexual orientation, gender identity, political affiliation or belief, genetic information, or other legally protected status. The University's commitment to nondiscrimination, including against sex discrimination, applies to students, employees, and applicants for admission and employment. Contact information, applicable laws, and complaint procedures are included on UA's statement of nondiscrimination available at www.alaska.edu/nondiscrimination.

Table of Contents

Unit Roster	6
Federal Scientists	6
University Staff.....	6
Unit Students	6
Unit Post-Doctoral Researchers.....	6
Unit Students Graduated This Year	6
Affiliated Students	7
Affiliated Students Graduated This Year	7
University Cooperators	7
Unit Coordinating Committee.....	8
Background.....	9
Land Acknowledgement.....	10
Statement of Direction	10
Message from the Unit Leader	11
Dr. Jeff Muehlbauer	11
Message from the Assistant Unit Leader – Wildlife	12
Dr. Megan Milligan.....	12
Unit Cost-Benefit Statements.....	13
In-Kind Support	13
Benefits	13
Courses Taught	13
Theses and Dissertations of Unit Graduate Students.....	13
Scientific and Technical Publications	13
Data and Code Releases.....	14
Presentations	14

Completed Fisheries Studies17

Zebra and Quagga Mussel Invasion Risk Project	17
Effects of Wildfire Disturbance on Fish Movement and Physiology in Boreal Streams	18
Alaska Forage Fish Data Compilation and Modeling for Risk Assessment and Resource Management in Oil and Gas Development Regions	19
Pink Arctic - Are Salmon Successfully Spawning in a North Slope River?	20

Ongoing Fisheries Studies21

Seaward Migration and Overwintering Habits of Dolly Varden in Northwestern Alaska Analyzed Using Otolith Microchemistry	21
Towards a Sustainable Future for Sport Black Rockfish Harvest in Southeast Alaska: Life History Changes, Surplus Production Models, and Fishery Knowledge.	22
Assessing Salmon Habitat and Shoreline Vulnerability to Recreational Boat Wakes in Big Lake, Alaska	23
The Influence of Stream Flow Patterns on Juvenile Salmon Foraging and Growth in Southeast Alaska	24
Climate Vulnerability of Aquatic Species to Changing Stream Temperatures and Wildfire Across the Yukon and Kuskokwim River Basins, Alaska	25
Evaluating Environmental DNA as a Complementary Tool for Estimating Salmon Abundance in the Yukon River Basin	26
Spatial and Decadal-Scale Temporal Changes in Water Chemistry and Macroinvertebrates in Central Alaska	27
The Effects of Water Temperature and Heat Stress on Juvenile Chinook and Coho Salmon Growth in the Deshka River Watershed.	28
Assessing Macroinvertebrate Community Response Following Stream Restoration to Guide Freshwater Conservation.....	29
Climate and Landscape Influences on Lake Vertical Structure in Southwest Alaskan National Parks.....	30
Rusting Rivers in Northwest Alaska Impacts on Stream Biota	31
Linking Macroinvertebrate Communities and Fish Bioenergetics in a Mined Subarctic Stream	32
Distributions and Ecology of Arctic and Pacific Lampreys in Alaska.....	33

Completed Wildlife Studies.....34

Investigating the Role of Juvenile Weight in Predicting Future Fitness in Northwestern Alaska Moose Populations	34
Movement and Habitat Use of Female Sea Otters (<i>Enhydra lutris</i>) In Kachemak Bay, Alaska	35
Small Mammal Disease Prevalence in Interior Alaska	36

Ongoing Wildlife Studies.....37

Evaluating Spatial-Capture-Recapture Models for Social Carnivores.....	37
Big Game Movement and Vital Rate Modeling	38
Factors Limiting Population Growth of Unhunted Dall Sheep	39
Investigating Aerial Insectivore Phenology as Related to Insect Abundance and Richness Within Boreal Alaska	40
Synthesizing Ungulate Movement and Migration Across Alaska	41
Developing a Multi-Species Monitoring Framework in Southeast Alaska	42
Black-Tailed Deer Density, Habitat Use, and Predator-Prey Dynamics.....	43
Human-Bear Conflict in Juneau, AK	44
Small Mammal Population Dynamics and Genetic Variation Prior to Development in Alaska National Parks	45
Distribution and Habitat Use of Nesting Surfbirds in the Steese National Conservation Area	46
Effects of Human Visitation on Bear Activity at Anan Observatory	47
Little Brown Bat Habitat Association Modeling & Geodatabase Development.....	48

Ongoing Ecological Studies49

Arctic Biogeochemical Response to Permafrost Thaw (ABRUPT)	49
--	----

List of Abbreviations50

Unit Roster

Federal Scientists

- Jeff Muehlbauer: Unit Leader–Fisheries
- Megan Milligan: Assistant Unit Leader–Wildlife
- *Vacant*: Assistant Unit Leader (x3)

University Staff

- Monica Armbruster: Fiscal Professional
- *Vacant*: Administrative Generalist

Unit Students

- Nate Cathcart, PhD Fisheries Student (Muehlbauer)
- John Hermus, MS Fisheries Student (Muehlbauer)
- Thomas House, MS Fisheries Student (Muehlbauer)
- Dakota Keller, MS Biological Sciences Candidate (Muehlbauer)
- Ian Kwit, MS Biological Sciences Candidate (Muehlbauer)
- Kayleigh McCarthy, MS Wildlife Biology and Conservation Candidate (Crimmins)
- Christie Osburn, MS Wildlife Biology and Conservation Student (Crimmins, Milligan)
- Rebecca Shaftel, PhD Fisheries Student (Falke)
- Sam Simon, MS Wildlife Biology and Conservation Student (Crimmins)
- Miles Spathelf, PhD Biological Sciences Student (Crimmins, Milligan)
- Sarah Swanson, PhD Biological Sciences Student (Crimmins)
- Sarah Trujillo, PhD Biological Sciences Student (Crimmins)
- Hannah Vincelette, MS Wildlife Biology and Conservation Student (Crimmins, Milligan)
- Eli Wilson, MS Fisheries Student (Muehlbauer)
- Lauren Yancy, MS Fisheries Student (Muehlbauer)
- Tony Zambito, MS Wildlife Biology and Conservation Candidate (Crimmins)

Unit Post-Doctoral Researchers

- Derek Arnold (Crimmins)
- Matt Dunkle (Muehlbauer)

Unit Students Graduated This Year

- Sara Henslee, MS Wildlife Biology and Conservation (Crimmins)
- Elizabeth Hinkle, PhD Fisheries (Falke)
- Tabitha Page, MS Wildlife Biology and Conservation (Crimmins)

Affiliated Students

- Margaret Harings, MS Fisheries (López and Schoen)
- Elizabeth Lindley, PhD Fisheries (Westley)
- Isaac Nyameke, PhD Fisheries (Westley)
- Joe Spencer, MS Fisheries (Seitz)
- Binta Wold, MS Wildlife Biology and Conservation (Brinkman)

Affiliated Students Graduated This Year

- Luke Porter, MS Wildlife Biology and Conservation (Breed)
- Lindsay Turner, MS Fisheries (Cunningham)

University Cooperators

- Peter Bieniek, IARC
- Greg Breed, DBW, IAB
- Todd Brinkman, DBW, IAB
- Morag Clinton, VetMed, Sitka Sound Science Center
- Shawn Crimmins, IAB
 - Former AKCFWRU Assistant Unit Leader – Wildlife, now Indiana Unit Leader
- Curry Cunningham, CFOS
- Arlo Davis, IARC
- Eugénie Euskirchen, DBW, IAB
- Jeff Falke, IAB
 - Former AKCFWRU Unit Leader, now Nevada Unit Leader
- Hélène Genet, IAB
- Brad Griffith, IAB
- Nina Hansen, VetMed
- Laura Conner, Vice Chancellor for Research, UAF
- Brian Smentkowski, Vice President for Academics, Students, and Research, UA Statewide
- J. Andrés López, CFOS
- A. David McGuire, IAB
- Charlene Stern, Acting Provost, UAF
- Caleb Robbins, Baylor University
- T. Scott Rupp, IARC, SNAP
- Will Samuel, IARC
- Erik Schoen, IARC
- Tobias Schwoerer, IARC
- Andy Seitz, CFOS
- Michael Sfraga, Acting Chancellor, UAF
- Deanna Strohm, CFOS
- Trent Sutton, CFOS
- Peter Westley, CFOS

Unit Coordinating Committee

- Toshihide Hamazaki – Chief Fisheries Scientist, Alaska Department of Fish and Game
- Chris Krenz – Wildlife Science Coordinator, Alaska Department of Fish and Game
- Kevin Whalen – Unit Supervisor, Cooperative Research Units, U.S. Geological Survey
- Diane O’Brien – Director, Institute of Arctic Biology, University of Alaska Fairbanks
- Tahzay Jones – Deputy Assistant Regional Director, Region 7, U.S. Fish and Wildlife Service
- Jen Newmark – Western Conservation Coordinator, Wildlife Management Institute

Background

This is the Annual Report for the Alaska Cooperative Fish and Wildlife Research Unit, highlighting activities for State of Alaska fiscal year 2025. The Unit engages in research on living natural resources for a variety of State and Federal agencies. As an unbiased research organization, the Unit provides information requested and funded by these agencies. When studies are completed, the agencies use the information to assist in their natural resource management efforts. Most of the research is conducted by graduate students, many of whom go on to work for the agencies upon graduation.

The Alaska Unit was established in 1950 for wildlife, with the addition of a separate fisheries unit in 1978, and a merger into today's Alaska Cooperative Fish and Wildlife Research Unit in 1991. Across this history, the Unit has provided three-quarters of a century of research dedicated to helping conserve and enhance the living natural resources of the State and the Arctic Region. The Unit is part of a larger and even older program, the U.S. Department of the Interior's Cooperative Research Unit Program. Established in 1935, Cooperative Research Units were created to fill the vacuum of wildlife management information and the shortage of trained wildlife biologists. In 1960, the Unit Program was formally sanctioned by Congress with the enactment of the Cooperative Units Act. Each unit is a partnership between the Ecosystems Mission Area of the U.S. Geological Survey, a host university, a state fish and game agency, the U.S. Fish and Wildlife Service, and the Wildlife Management Institute. Staffed by Federal scientists, Cooperative Research Units conduct research on renewable natural resource questions, participate in the education of graduate students destined to become natural resource managers and scientists, provide technical assistance and consultation to parties who have legitimate interests in natural resource issues, and provide continuing education for natural resource professionals. Presently, there are 43 Cooperative Research Units in 41 states conducting research on virtually every type of North American ecological community. The Program is staffed by more than 100 PhD scientists who advise as many as 675 graduate student researchers per year.

Land Acknowledgement

We acknowledge the Alaska Native nations upon whose ancestral lands our campus resides. The University of Alaska Fairbanks' Troth Yeddha' Campus, where the Unit is located, is on the traditional lands of the Dena people of the lower Tanana River. Our research throughout Alaska takes place on the unceded lands of the Haida, Tsimshian, ƛingít/Tlingit, Eyak, Athabaskan, Yup'ik, Cup'iq, Sugpiaq/Alutiiq, Unanga/Aleut, and Iñupiaq peoples. We respect and appreciate their past and present stewardship of these lands and waters.

Statement of Direction

The research program of the Unit will be aimed at understanding the ecology of Alaska's fish and wildlife, evaluating impacts of land use and development on these resources, and relating effects of social and economic needs to production and harvest of natural populations.

In addition to the expected Unit functions of graduate student training/instruction and technical assistance, research efforts will be directed at problems of productivity, socioeconomic impacts, and perturbation on fish and wildlife populations, their habitats, and ecosystems. Fisheries research will emphasize water quality, habitat characteristics, and life history requirements of northern fish populations. Wildlife research will focus on the ecology of northern birds and mammals and their habitats. Unit research will also be directed at integrated studies of fish and wildlife at the ecosystem level.

Message from the Unit Leader

Dr. Jeff Muehlbauer

It has been an eventful and in many ways challenging year for AKCFWRU. However, reflecting on the past twelve months my lasting emotion is one of gratitude:

I am especially grateful that we have successfully recruited a new Assistant Unit Leader – Wildlife, Dr. Megan Milligan, who started in early 2025. It has been rewarding to see Dr. Milligan start making connections statewide and build her own research lab focusing on wildlife. I'm confident future annual reports will be stuffed with fascinating research from her and her incoming students.

I'm also thankful for the continued support of our cooperators, and to have gotten to know many of them much better as we worked closely in the Assistant Unit Leader recruitment. Their unwavering dedication to the mission of the Unit has also been on full display this year, for which I am deeply grateful. My vision is for the Unit to be highly visible and improve its standing as the face of cutting-edge fisheries and wildlife research in the state of Alaska. A silver lining of this year of governmental uncertainty has been that—through the dedicated efforts of our cooperators—many people around UAF, the state, and even the nation have become much more aware of AKCFWRU and its mission. My humble promise is to continue to do my best to further that mission and continue to earn your cooperator support.

For my research program in Fisheries, I have been excited to see my first cohort of graduate students make excellent progress toward their degrees, with several planning thesis defenses within the next few months. I have also been deliberately trying to bring my lab's research into increased thematic focus, with several students now carrying out research in or adjacent to stream restoration as well as species responses to freshwater environmental change. At the same time, I do my best to be reflexive and adaptable to working on new projects on varied topics as partner needs arise, which I value as an important part of the mission of the Unit. In all these I am continually impressed by the insight and inquisitiveness of the personnel in my research group who carry out the bulk of this research; they challenge me constantly and I am grateful for the extraordinary privilege of working with them.

Looking forward, I remain hopeful that AKCFWRU will be able to recruit additional Assistant Unit Leaders, and I will continue to push for this whenever a reasonable opportunity should arise. Regardless, I am confident in the direction of my research group in fisheries and freshwater ecology, and in Dr. Milligan's parallel but necessarily nascent efforts in wildlife ecology. As evidenced by the breadth of research in the following pages of this report, I believe AKCFWRU continues to be a productive force in fisheries and wildlife research, and that our future remains bright.



Message from the Assistant Unit Leader – Wildlife

Dr. Megan Milligan

Greetings!

This past January, I joined AKCFWRU as the new Assistant Unit Leader – Wildlife and I am both excited and humbled by the opportunity to join the Unit. I cannot think of a better place to pursue important applied research, mentor students, and serve the needs of cooperators. I have already had the pleasure of learning a great deal from Dr. Muehlbauer's experience and wisdom, and I look forward to continuing to learn from and work with the Unit's many collaborators going forward.

As the newest member of the Unit, I have relished learning about all of the recent and ongoing wildlife research being done in Alaska. I envision the Unit continuing to operate as a vibrant component of this research community, both serving our cooperators and connecting researchers across the state. As a first step in building my research program here, I have spent the last several months getting to know many of our collaborators and learning more about their research needs and interests. I look forward to connecting with those of you who I have not had the pleasure of meeting. One of the key aspects that drew me to this position was the chance to work collaboratively on applied research questions that were of direct relevance to our cooperators. For this reason, I have greatly appreciated these opportunities to learn more about what different scientists around the state perceive as the most pressing research needs or issues. In the coming year, I hope to continue to increase the Unit's connections with scientists and cooperators across Alaska to explore new research projects and collaborations.

I was also drawn to this position by the opportunity to work directly with graduate students. I enjoy mentoring students and helping them build both the critical thinking and technical skills to help them become successful wildlife professionals. I am excited to be currently co-advising three students with Shawn Crimmins and I look forward to recruiting more graduate students in the coming year.

I am originally from New Mexico, but I previously had the pleasure of living in several parts of Alaska and I am delighted to be back. I feel incredibly fortunate to have a job that allows me to both work and play in such an amazing place. I am also thankful to have the opportunity to work with so many scientists who care deeply about both Alaska's wildlife and wild places. I look forward to collaborating with many of you in the future and contributing to our shared goals regarding Alaska's wildlife and landscapes.



Unit Cost-Benefit Statements

In-Kind Support

In-kind support, usually operational support of field activities, is critical to the success of the Alaska Cooperative Fish and Wildlife Research Unit. Although the monetary value of this support is not known, a listing of this assistance is provided for each project in this report.

Benefits

- Students Graduated (advised by Unit faculty): 3
- Scientific and Technical Publications: 6
- Data and Code Releases: 4
- Presentations: 26

Courses Taught

Jeff Muehlbauer: Aquatic Food Web Ecology (UAF, 3 credits, Fall 2024)

Jeff Muehlbauer: Readings in Stream Ecology (UAF, 1 credit, Spring 2025)

Theses and Dissertations of Unit Graduate Students

Henslee, S.R., 2024, Effects of juvenile mass on reproduction and calf survival in a low-density moose population: Fairbanks, Alaska, USA, University of Alaska Fairbanks, M.S., 75 p.

Hinkle, E.A., 2025, Post-fire riverscapes: Exploring the ecological and behavioral responses of stream communities to wildfire in boreal streams: Fairbanks, Alaska, USA, University of Alaska Fairbanks, M.S., 160 p.

Page, T., 2025, Molecular survey of *Leptospira* and *Bartonella* in Alaskan voles: Fairbanks, Alaska, USA, University of Alaska Fairbanks, M.S., 51 p.

Scientific and Technical Publications

Cathcart, C.N., Falke, J.A., Fox, J., Henszey, R. and Lininger, K., 2024, Multiscale processes drive formation of logjam habitats and use by juvenile Chinook salmon across a boreal stream network in Alaska: River Research and Applications v. 41, p. 593–608. DOI: 10.1002/rra.4387

Fitzgerald, K.A., Bellmore, J.R., Fellman, J.B., Cheng, M.L.H., Boyles-Muehleck, N., Delbecq, C.E., and Falke, J.A., 2024, Juvenile coho salmon growth differences track biennial pink salmon spawning patterns: Freshwater Biology, v. 69, p.1583–1595. DOI: 10.1111/fwb.14328.

Merems, J.L., Brose, A.L., Price Tack, J., Crimmins, S., and Van Deelen, T.R., 2024, Scale-dependence in elk habitat selection for a reintroduced population in Wisconsin, USA: Ecology and Evolution, v. 14, e70346. DOI: 10.1002/ece3.70346

Meyerpeter, M.B., Coates, P.S., Milligan, M.C., Prochazka, B.D., Lazenby, K.D., Abele, S., Tull, J., Miller, K., Kolar, J., Mathews, S.R., Dahlgren, D.K., and Delehanty, D.J., 2025, Conservation translocation immediately reverses decline in imperiled sage-grouse populations: Biological Conservation v. 304, 110986. DOI: 10.1016/j.biocon.2025.110986

Milligan, M.C., Coates, P.S., O'Neil, S.T., Brussee, B.E., Chenaille, M.P., Friend, D., Steele, K., Small, J.R., Bowden, T.S., Kosic, A.D., and Miller, K., 2025, Integrating multiple indices of habitat quality to inform habitat management for a sagebrush indicator species: *Ecological Informatics* v. 90, 103228. DOI: 10.1016/j.ecoinf.2025.103228

Strohm, D.D., Sergeant, C.J., Paul, J.D., and Falke, J.A., 2025, Streamflow regime characterization in the changing boreal ecosystem: wildfire impacts from stream-to-regional scales: *Science of the Total Environment*, v. 991, 179770. DOI: 10.1016/j.scitotenv.2025.179770

Data and Code Releases

Coates, P.S., Milligan, M.C., O'Neil, S.T., and Brussee, B.E., 2025, Code to examine habitat selection and survival of greater sage-grouse in Nevada and northeastern California: U.S. Geological Survey Software Release. DOI: 10.5066/P13PTWQA

Coates, P.S., Meyerpeter, M.B., Milligan, M.C., Prochazka, B.G., Mathews, S.R., and Chenaille, M.P., 2025, Survival of greater sage-grouse nests and broods following translocation in California, Wyoming, and North Dakota, 2002-2019: U.S. Geological Survey Data Release. DOI: 10.5066/P1H2MBHK

Fitzgerald, K.A., and Falke, J., 2024, Juvenile coho salmon growth differences track biennial pink salmon spawning patterns: U.S. Geological Survey Data Release. DOI: 10.5066/P1PWJM7G

Toskey, E.K., Matsuoka, T., Kiffney, P.M., Naman, S.M., Liermann, M.C., Bailey, J., Beauchamp, D.A., Bellmore, J.R., Binckley, C.A., Blumenshine, S., Caldwell, T.J., Carey, M.P., Claeson, S., Clawson, C., Cox, R., Cromwell, K.J., Danehy, R.J., Dodrill, M.J., Duda, J.J., Fellman, J.B., Grimm, A.Z., Hallbert, T.B., Jordan, C.E., Keeley, E.R., Kennedy, B.P., Kennedy, T.A., Kohler, A.E., Lowery, E.D., Macneale, K., Martens, K.D., Morley, S.A., Muehlbauer, J.D., Neuswanger, J.R., O'Dowd, A., Pneh, S., Power, M., Rhoades, S.J., Rosenfeld, J., Rossi, G., Sanderson, B.L., Shelby, K.K., Starky-Owens, T., Thompson, J.N., Volk, C., Williamshen, J.S., and Wisseman, R.W., 2025, DRIFteRS (DRift Invertebrates For salmonids in River Systems): A dataset of drift invertebrate densities in streams and rivers across western North America, 1997–2024 ver 1, Environmental Data Initiative. DOI: 10.6073/pasta/234b23f08b48458e7d3910045ebacd9f

Presentations

Brussee, B. E., Coates, P.S., Milligan, M.C., O'Neil, S.T., Chenaille, M.P., and Friend, D.A., 2025, Using remotely sensed fractional cover data to inform wildlife habitat suitability and management: Annual Meeting of the Society for Range Management, Spokane, WA, USA.

Coates, P.S., Milligan, M.C., Weise, C.L., Friend, D.A., Lundblad, C.G., Severson, J.P., and Olsen, A.C., 2025, Greater sage-grouse and pinyon-juniper: an overview of the science: Annual Meeting of the Society for Range Management, Spokane, Washington, USA.

Dempsey, S., Zavoico, V.S., Aderman, A.A., and Crimmins, S.M., 2025, Lifetime resource selection by an expanding subarctic moose population: behavioral links to individual demography: American Society of Mammalogists Annual Conference, West Lafayette, IN, USA.

- Falke, J., Schoen, E., Paul, J., Strohm, D., and Hinkle, E., 2025, Implications of wildfire on stream thermal regimes and juvenile salmon growth in interior Alaska boreal streams: Western Division American Fisheries Society Annual Meeting, Westminster, Colorado.
- Falke, J., Schoen, E., Paul, J., Strohm, D., and Hinkle, E., 2025, Implications of wildfire on stream thermal regimes and juvenile salmon growth in interior Alaska boreal streams: Society for Freshwater Science Annual Meeting, San Juan, Puerto Rico.
- Harings, M.A.B., Schoen, E., Kamermans, B. Reece, K., Falke, J. Matter, A., McKenna, B. Farnham, N., Walter, S.T., and López, J.A., 2025, Evaluating environmental DNA as a complementary technique for assessing Yukon River salmon: Alaska Chapter of the American Fisheries Society Annual Meeting, Ketchikan, Alaska.
- Hermus, J.W., Muehlbauer, J.D., Rinella, D.J., von Biela, V.R., and Falke, J.A., 2025, The effects of heat stress on juvenile Chinook and Coho Salmon growth in the Deshka River: Alaska Chapter American Fisheries Society Annual Meeting, Ketchikan, Alaska.
- Hermus, J.W., Muehlbauer, J.D., Rinella, D.J., von Biela, V.R., and Falke, J.A., 2024, The effects of heat stress on juvenile Chinook and Coho Salmon growth in the Deshka River: Mat-Su Salmon Science and Conservation Symposium, Palmer, Alaska.
- Hinkle, E., Wooller, M., Westley, P., Spaleta, K., and Falke, J., 2025, Movement strategies of Arctic grayling within a fire-impacted riverscape: Alaska Chapter of the American Fisheries Society Annual Meeting, Ketchikan, Alaska.
- House, T.H., and Muehlbauer, J.D., 2025, Before we restore: examining energetic budgets of juvenile salmonids in a mining-impacted stream: Alaska Chapter American Fisheries Society Annual Meeting, Ketchikan, Alaska.
- House, T.F., and Muehlbauer, J.D., 2025, Pre-restoration assessment for juvenile salmon growth in a mined Alaskan stream: Society for Freshwater Science Annual Meeting, San Juan, Puerto Rico.
- Keller, D.E., and Muehlbauer, J.D., 2025, Changes in macroinvertebrate communities following different stream restoration approaches: Alaska Chapter American Fisheries Society Annual Meeting, Ketchikan, Alaska.
- Lindenauer, N., Coates, P.S., Milligan, M.C., Mathews-Sanchez, S.R., Meyerpeter, M.B., and Patricelli, G.L., 2025, Augmenting translocated sage-grouse broods and guiding future release locations: Annual Meeting of the Western Section of The Wildlife Society, Visalia, California, USA.
- Milligan, M.C., Coates, P.S., O'Neil, S.T., Brussee, B.E., McLachlan, M., Aldridge, C.L., Wann, G.T., and Small, J.R., 2025, Estimating spatial variation in greater sage-grouse space use relative to leks to inform regional buffers and surface use designations: Annual Meeting of the Society for Range Management, Spokane, WA, USA.
- Muehlbauer, J.D., 2024, "Bug Flows" to benefit aquatic food webs in Grand Canyon: Fisheries Seminar Series, University of Alaska Fairbanks, Fairbanks, Alaska.

- Muehlbauer, J.D., 2024, Environmental flows for food webs and fishes in Grand Canyon: School of Fisheries and Aquatic Sciences Seminar Series, University of Washington, Seattle, Washington.
- Muehlbauer, J.D., 2025, What's 250 microns between friends? How choice of sampling gear affects food web data: Alaska Chapter American Fisheries Society Annual Meeting, Ketchikan, Alaska.
- Page, T., Hansen, C., Crimmins, S., 2024, Voles as hosts to emerging bacterial diseases in Alaska: International Conference of the Wildlife Disease Association, Canberra, Australia.
- Shaftel, R., Bellmore, R., Cunningham, C., Schoen, E., and Falke, J., 2025, Using life cycle models to understand effects of extreme streamflow and water temperature events on Yukon River basin salmon populations: Alaska Chapter of the American Fisheries Society Annual Meeting, Ketchikan, Alaska.
- Toney, B., Bellmore, J., Fellman, J., Falke, J., and Cunningham, C., 2025, Impacts from hydrologic variability on juvenile salmonid movement and growth potential in a Southeast Alaskan watershed: Alaska American Fisheries Society Annual Meeting, Ketchikan, Alaska.
- Wesner, J., Larsen, S., Kraus, J.M., Muehlbauer, J.D., Kopp, D., Schultz, R., Wolfram, J., Álvarez-Martínez, J.M., and Grigoropoulou, A., 2025, Mapping the global flux of biomass and contaminants from rivers by emerging aquatic insects: Society for Freshwater Science Annual Meeting, San Juan, Puerto Rico.
- White, K.M., Kynoch, M.C., Savory, G., Jochum, K.A., and Crimmins, S.M., 2025, Deciduous trees facilitate while shrubs inhibit little brown bat (*Myotis lucifugus*) distribution in interior Alaska: American Society of Mammalogists Annual Conference, West Lafayette, IN, USA.
- Wilson, E.E., and Muehlbauer, J.D., 2025, Boats and beach spawners: impacts of boat wakes on shoreline erosion and an assessment of shoreline habitat use by Sockeye Salmon at Big Lake, AK: Alaska Chapter American Fisheries Society Annual Meeting, Ketchikan.
- Wilson, E.E., Muehlbauer, J.D., 2024, Effects of watercraft wakes on shoreline erosion, and potential impacts for salmon at Big Lake, Alaska: Mat-Su Salmon Science and Conservation Symposium, Palmer, Alaska.
- Yancy, L.E., and Muehlbauer, J.D., 2025, Impacts of rusting rivers on stream biota in the Wulik River drainage, Northwest Alaska: Alaska Chapter American Fisheries Society Annual Meeting, Ketchikan, Alaska.
- Yancy, L. E., and Muehlbauer, J.D., 2025, Long-term biomonitoring data reveals impacts of rusting rivers in Northwest Alaska: American Fisheries Society Annual Meeting, San Antonio, TX, USA.

***Note:** Reports are listed as Completed or Ongoing in the categories of Fisheries, Wildlife, or Ecological Studies, in that order. It is recognized that these distinctions are somewhat artificial. A list of abbreviations appears on the final page of the report. For completed studies, the abstract of the student's thesis or dissertation is provided where possible.

Completed Fisheries Studies

Zebra and Quagga Mussel Invasion Risk Project

Staff Scientist: Deanna Strohman (CFOS)

PI: Tobias Schwoerer and Erik Schoen (IARC)

Funding Agency: ADF&G



Technician Connor Ford collecting water samples at Birch Lake, Alaska

Invasive zebra and quagga mussels have caused major economic and ecological damage to lakes, rivers, fisheries, and human infrastructure in the Lower 48 US states as well as parts of Europe and Canada. While these species have not yet established populations in Alaska, they have been detected on used boat hulls crossing the border on the Alcan Highway. This project aims to expand Alaska's water quality data set to identify lakes that are at risk of invasive mussel invasion based on water chemistry and other characteristics, to prevent their establishment in our waters. Our specific objectives for the 2024 field season were to: 1) identify road accessible lakes across southeast Alaska lacking calcium data, and 2) partner with local nonprofit, tribal, and federal agencies working in these areas to collect water samples for calcium analysis. Our team at UAF is working with state, federal, Tribal, and non-profit partner organizations to collect water samples from high priority lakes in Interior and Southcentral Alaska during the 2024 field season.

A total 105 water samples were collected from twenty-five lakes throughout Alaska. Calcium concentrations ranged from 0.9 ppm to 60.9 ppm with an average of 14.47 ppm, a median of 8.29 ppm, and standard deviation of 14.2 ppm. Of the 25 lakes sampled, eight fell within the high-risk range for dreissenid mussel invasion (>20 ppm Ca), five were classified as moderate risk (8–20 ppm Ca), and twelve were in the low-risk range (<7.9 ppm Ca).

The data collected from this project will enhance our statewide calcium dataset, providing a better understanding of which lakes are at the highest risk of mussel invasion and enabling managers to prioritize prevention and management efforts more effectively.

Effects of Wildfire Disturbance on Fish Movement and Physiology in Boreal Streams

Student Investigator: Elizabeth Hinkle, PhD Fisheries

PI: Jeff Falke

Funding Agency: DoD SERDP (RWO 227)

In-Kind Support: ADF&G

Wildfire is the dominant disturbance in boreal forest ecosystems, and its frequency and severity are increasing with climate change. This dissertation examines how wildfire affects boreal stream habitats, food webs, and fish movement across spatial and temporal scales in interior Alaska. In Chapter 2, I compared burned and unburned streams 10 to 15 years post-fire using habitat survey, water chemistry, and aquatic community composition data. Burned sites had marginally significant lower canopy cover (-19.6% , $p = 0.095$), coarser substrate ($+23.7$ mm, $p = 0.102$), and higher macroinvertebrate abundance (626 ± 361 vs. 331 ± 349 individuals/m², $p = 0.03$); fish biomass was more than twice as high in burned sites. In Chapter 3, I combined stable isotope analysis ($\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values) with simulations from the Aquatic Trophic Productivity (ATP) model to evaluate how wildfire alters food web structure and productivity for up to 50 years post-fire. Field observations revealed that burned streams had greater aquatic macroinvertebrate diversity, longer food chains ($\delta^{15}\text{N}$ range = 6.2% vs. 3.4%), and lower $\delta^{13}\text{C}$ values in primary producers, suggesting increased algal energy inputs to food webs. ATP model outputs supported these patterns, predicting persistent increases in macroinvertebrate biomass and fish production for decades following fire. Bayesian isotope mixing models further indicated that juvenile Chinook salmon (*Oncorhynchus tshawytscha*) and Arctic grayling (*Thymallus arcticus*) in burned streams relied more on algal-based food resources and exhibited broader niche widths. Chapter 4 applied a spatial stream network model and strontium isoscape to estimate where Arctic grayling reared and moved throughout life. Approximately 20% of modeled fish locations occurred in burned areas, which is roughly proportional to their availability ($\sim 23\%$ of the basin), suggesting no strong preference or avoidance at the basin scale throughout the lifetime of individual fish. Larger, mature fish with higher dry mass were disproportionately associated with burned habitats. Life-history clusters characterized by long-distance movement were more common in burned sites, suggesting that post-fire habitats may provide key foraging opportunities within the boreal riverscape. Overall, my dissertation integrated empirical field data, modeling approaches, and spatial analyses to provide a broad assessment of how wildfire influences the structure and function of boreal freshwater ecosystems.

Alaska Forage Fish Data Compilation and Modeling for Risk Assessment and Resource Management in Oil and Gas Development Regions

Student Investigator: Lindsay Turner, MS Fisheries

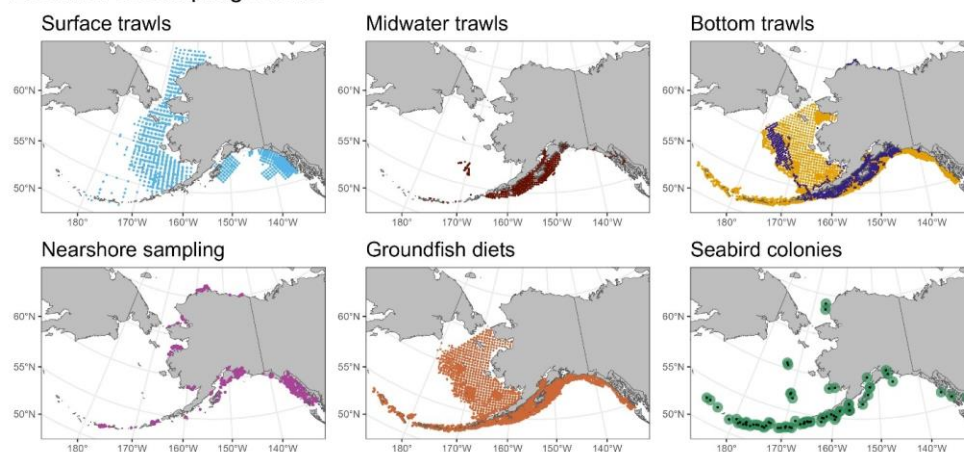
PI: Curry Cunningham, CFOS

Funding Agency: BOEM (RWO 245)

In-Kind Support: USGS

Forage fish are energy rich species that play a crucial role in the diets of seabirds, marine mammals, and commercially important fish species in Alaska. However, data limitations present a challenge for quantifying the spatial and temporal dynamics of forage fish, and their past responses to ecosystem change. To address key knowledge gaps surrounding the population and spatial dynamics of forage fish within the waters surrounding Alaska, we have compiled and published the Alaska Forage Fish Database (AFFD) as part of a USGS Alaska Science Center data release (<https://www.usgs.gov/data/alaska-forage-fish-database-affd>). Through the collaborative effort of multiple agencies who have led past data collection and survey efforts, we have aggregated 10 data sources (57 different projects) describing forage fish occurrence and catch rate information, from independent fisheries surveys and the diets of seabirds and groundfish that index forage fish. The AFFD is the largest compilation of forage fish data in Alaska to date, with 461,449 forage fish samples from 105,420 unique sites, covering the years 1953-2023. Using the AFFD, we estimate the trends in occurrence and distribution of key forage species (Pacific sand lance, Pacific herring, and Pacific capelin) in the Gulf of Alaska and Bering Sea. We implement models fit to single and multiple types of data (e.g., fisheries surveys, groundfish diets, seabird diets) to assess changes in forage fish distribution in response to decadal climate regimes. The integration of multiple sources of occurrence information from fisheries surveys and predator diets provides a basis for expanding the spatial and temporal coverage of data and facilitating a more comprehensive understanding of forage fish distribution in Alaska and responses to past environmental change.

Location of sampling events



Spatial coverage of the six primary sampling methods: a) surface trawls (light blue), b) midwater trawls (dark red), c) bottom trawls (small-mesh in dark blue, large-mesh in yellow), d) nearshore sampling (purple), e) locations of groundfish stomach content collection (orange), and f) seabird colonies (green circle with black point). All individual sampling events are included as points, with the exception of seabird colony data.

Pink Arctic - Are Salmon Successfully Spawning in a North Slope River?

Student Investigator: Elizabeth Lindley, PhD Fisheries

PI: Peter Westley, CFOS

Funding Agency: USFWS (RWO 252)

In-Kind Support: AKSG, NSF Tamamta, CFOS Wakefield Chair



Peter Westley retrieving temperature logger from Anaktuvuk River spawning site, September 2024. Photo credit: Stephanie Carlson

Across the rapidly warming Arctic, life is on the move. Species once limited to subarctic regions are pressing northward, reshaping food webs and human relationships with the land and sea. Among the most striking of these redistributions are Pacific salmon (*Oncorhynchus* spp.), long associated with temperate and subarctic rivers but now appearing with increasing regularity in the far north. Indigenous Peoples have harvested salmon from nearshore waters of the North Slope for over a century, yet observations and harvests have grown noticeably in recent decades. Whether these fish represent transients, or the

beginnings of self-sustaining Arctic populations is a question of ecological and cultural importance. To explore this possibility, we focused aerial surveys on tributaries of the Colville River known to support anadromous Dolly Varden (*Salvelinus malma*), whose spawning needs closely parallel those of salmon. Aerial surveys in 2023 revealed spawning Chum Salmon (*O. keta*) and, for the first time, the presence of Sockeye Salmon (*O. nerka*). At these sites, we placed in-gravel temperature loggers to track incubation conditions through the Arctic winter. Retrieved in September 2024, the results from these novel records were striking. Temperatures approached but never fell below 0°C, providing thermal regimes suitable for embryo survival. When applied to developmental models, these data predicted hatching and emergence the following August—evidence that salmon can complete a critical life stage under Arctic conditions. This finding suggests that redistribution may extend beyond seasonal strays: the Arctic holds the capacity to carry the next generation of salmon.

Ongoing Fisheries Studies

Seaward Migration and Overwintering Habits of Dolly Varden in Northwestern Alaska Analyzed Using Otolith Microchemistry

Student Investigator: Joseph Spencer, MS Fisheries

PI: Andy Seitz, CFOS

Funding Agency: ADF&G Region 3 (Sport Fish Base), USFWS OSM

In-Kind Support: ADF&G, USFWS



Packrafting the upper Noatak River, Alaska.

Diadromous Dolly Varden are an important subsistence resource in northwestern Alaska. They feed in the ocean during ice-free months and rear, spawn, and overwinter in rivers. Frequent inter-river movements for overwintering and spawning are suspected, but life history and migratory strategies of populations across the region are not well understood. Movements among rearing, spawning, and overwintering habitats expose Dolly Varden to numerous subsistence, commercial, and sport fisheries.

Knowledge of migratory strategies is

needed to fully understand population dynamics, exposure to harvest, and impacts of climate change. The objective of this study is to elucidate the migration strategies of Dolly Varden populations using otolith microchemistry. Between 2021 and 2024, we collected and analyzed otoliths from Dolly Varden captured ($n=295$) in spawning areas of northwestern Alaska, from subsistence fisheries ($n=91$) in Kivalina and Noatak villages, and from bycatch in the commercial salmon fishery ($n=31$) in Kotzebue. Strontium concentrations measured by laser ablation along the growth axis of otoliths were used to determine the age at first seaward migration, frequency of seaward migrations, and movement between overwintering sites. Results indicate that Dolly Varden in the region typically smolt at age 3 and undergo 3–4 ocean migrations prior to spawning. Among spawners, 90.5% skipped at least one ocean migration during their lifetime, and repeat spawners comprise an average of 29% of the spawning population. This research will inform management of fisheries in the region and establish information about life history and migratory strategies to track changes in a rapidly warming Arctic.

Towards a Sustainable Future for Sport Black Rockfish Harvest in Southeast Alaska: Life History Changes, Surplus Production Models, and Fishery Knowledge.

Student Investigator: Isaac Nyameke, PhD Fisheries

PI: Peter Westley, CFOS

Funding Agency: ADF&G Region 1 (Sport Fish Base), NSF Tamamta & UAF Troth Yeddha'

Since the introduction of United States federal and Alaska statewide fisheries regulations in the early 2000s such as bag limits, seasonal closures, and gear restrictions on Pacific salmon and halibut aimed at controlling fishing pressure and ensuring sustainability, sport fishing effort has increasingly shifted toward black rockfish in Southeast Alaska. As a result, annual sport harvest of black rockfish has grown from approximately 200 tons in 2004 to more than 250 tons in 2024. This rapid increase has raised concerns among fishermen and scientists regarding the long-term sustainability of Southeast Alaska black rockfish stocks.

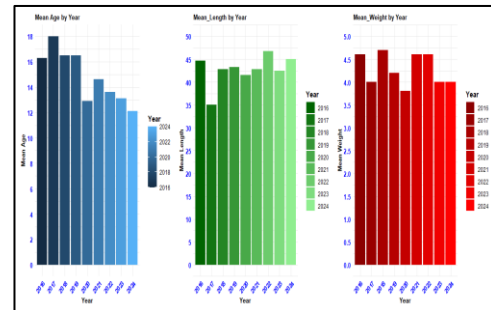


Figure 1. Annual temporal trends in mean age (years, left), length (cm, middle) and weight (kg, weight) across years.

This research is being conducted as part of the Alaska Department of Fish and Game (ADF&G) Statewide Rockfish Initiative (SRI). The overall objectives are twofold: (1) to quantify long-term trends in body size and age of Southeast Alaska black rockfish across nine coastal communities, using length, weight, and age data collected exclusively from sport fishing boats between 2016 and 2024, and (2) to evaluate the current stock status of Southeast Alaska black rockfish by comparing the performance of five data-limited assessment models.

Our preliminary results, based on generalized additive models (GAMs), indicate an 8.7% decline in mean length and a 24% decline in mean weight, with maximum observed age decreasing from 16 years in 2016 to 11 years in 2024, although the trend is not strictly linear (Figure 1). Older, larger females are increasingly underrepresented in the current catch whereas male size structure remains stable. Recent catches were found to be dominated by younger age classes as well indicating age and size truncation.

The Kobe plot (Figure 2) suggests that the Southeast Alaska black rockfish stock is near maximum sustainable yield but not yet overfished. The Bayesian state-space Schaefer biomass production model estimated a maximum sustainable yield threshold of 385 tons, indicating that harvests above this level may place the stock at risk of becoming overfished. Our study provides scientific evidence to guide management decision of long-term sustainability of sport black rockfish harvest in Southeast Alaska.

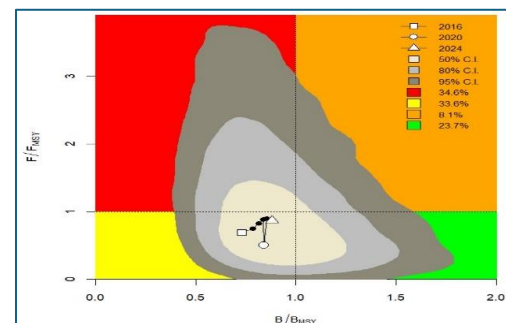


Figure 1. Kobe plot showing the stock status of Southeast Alaska black rockfish relative to biomass (B/B_{MSY}) and fishing mortality (F/F_{MSY}) reference points from 2016 to 2024.

Assessing Salmon Habitat and Shoreline Vulnerability to Recreational Boat Wakes in Big Lake, Alaska

Student Investigator: Eli Wilson, MS Fisheries

PI: Jeff Muehlbauer

Funding Agency: ADF&G RTS (Sport Fish Base), USGS SSP (RWO 255), UAF URSA

In-Kind Support: ADF&G, USFWS



Big Lake drone view. Photo credit: Eli Wilson.

The Matanuska-Susitna Valley is one of the fastest growing regions of Alaska in terms of population, which makes understanding the impacts of various land uses crucial to addressing current and future conflict between user groups. The Big Lake drainage supports the only road accessible personal use Sockeye Salmon fishery in Upper Cook Inlet, a crucial resource for local residents, which is dependent on system

health and capacity to support the species. In order to promote the conservation and rehabilitation of shoreline habitats key to ecosystem function, ADF&G has a long-standing partnership with USFWS to provide both funding and expertise to landowners willing to pursue fish-friendly, bioengineered approaches to streambank and lakeshore stabilization rather than hard infrastructure. Partners have reported increased concerns from landowners considering participation in this program over the years due to perceived increases in the size and frequency of boat wake events affecting shorelines, yet there are no quantitative data to pair with these anecdotal reports. Over the summer of 2024 we collected data on shoreline erosion rates, wake activity, and the spawning distribution of beach spawning Sockeye Salmon in the system. We found that for our nine erosion pin monitoring sites, season long mean erosion ranged from 7.7–91.5 mm. Mean two-week erosion across all sites and pins was 5.3 mm. At our five wake monitoring sites we documented large differences in the number of wakes detected impacting the shoreline at the various sites; however, we did not find the number of detected wakes to have a significant correlation with pin level erosion. Our spawning surveys identified six beach spawning areas utilized by Sockeye Salmon, which had not been formally documented prior to this study and confirmed that most spawning occurred in substrate dominated by pebbles and cobbles.

The Influence of Stream Flow Patterns on Juvenile Salmon Foraging and Growth in Southeast Alaska

Student Investigator: Blake Toney, MS Fisheries

PI: Curry Cunningham

Funding Agency: USGS AK CASC (RWO 244)

In-Kind Support: USFS PNWRS, UAS, ACRC



UAS Undergraduate researcher, Amelia Vegors, and UAF Graduate student, Blake Toney, tagging juvenile salmonids in Montana Creek near Juneau, AK. Photo credit: Lele Fujimoto.

Climate driven shifts in Southeast Alaska's hydrology are altering the freshwater conditions critical to wild salmon populations. Climate models project more frequent and intense summer precipitation events, resulting in extreme flow conditions in salmon-bearing streams. These changes may disproportionately affect Pacific salmonid species such as juvenile coho salmon and Dolly Varden char that rear in freshwater for extended periods before migrating to the ocean. Our study investigates the role of small, second-order tributaries to provide important habitat and foraging opportunities during high-flow events in salmon-bearing river systems.

Previous research suggested that tributaries

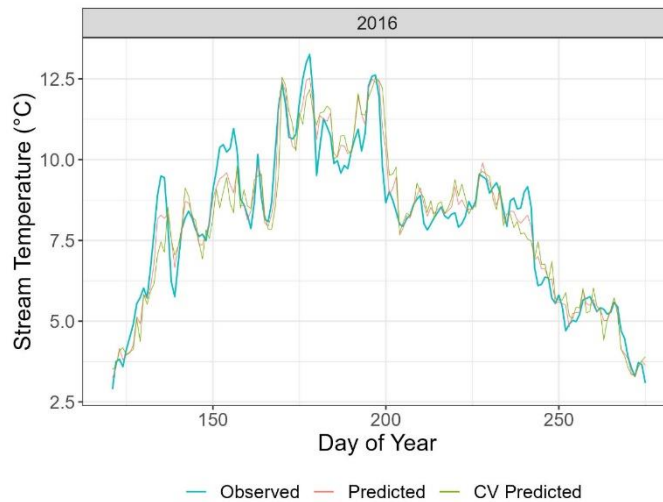
can serve as *high-flow refugia* by offering juvenile salmonids shelter from the energetic cost of mainstem conditions during peak discharge. However, increased turbidity and accelerated water velocity during high-flow events complicates active fish sampling, leaving the ecological value of intermittently accessible tributary habitats largely unquantified in coastal Southeast Alaskan streams. We use Passive Integrated Transponder (PIT) tags and portable antenna arrays to monitor juvenile coho and Dolly Varden movement between two perennial second-order tributaries and adjacent mainstem habitats and identify whether habitat use patterns influence growth potential. We complement movement data with diet sampling under a high frequency sampling design that targets variability in flow events to identify whether differences in food subsidies are responsible for variable growth potential among fish exhibiting divergent habitat use strategies. By integrating movement, diet, and growth data, our study aims to inform land management practices by quantifying the contribution of tributary habitats to population-level resilience under changing hydrologic regimes.

Climate Vulnerability of Aquatic Species to Changing Stream Temperatures and Wildfire Across the Yukon and Kuskokwim River Basins, Alaska

Student Investigator: Rebecca Shaftel, PhD Fisheries

PI: Jeff Falke

Funding Agency: USGS AK CASC (RWO 246)



Observed and predicted stream temperatures for Chena River from 2016.

Alaska is experiencing climatic change faster than any other area of the United States, but across the state, comprehensive environmental monitoring is logistically difficult and expensive. During the record heat of 2019, adult salmon mortalities were observed in many locations. In the mainstem Yukon, mortalities were closely linked to temperatures that have been shown to induce heat stress in Chinook Salmon. There is growing concern that warming stream temperatures and extreme events may be negatively impacting adult salmon, while effects on other life stages or other fish species are unknown due to

limited monitoring data and a lack of stream temperature information. The objective of this study is to compare modeling methods (including machine learning) for predicting stream temperatures under current and future climate scenarios across the Yukon and Kuskokwim River basins (YKRB) at the stream reach (~ 1 km) scale. The models will be derived from gridded climate products that include air temperatures and streamflow coupled with future climate model projections. Recently developed stream temperature models for 33 monitoring sites at Chinook Salmon migration routes and spawning tributaries across the YKRB achieved cross-validated prediction errors less than 1°C using similar inputs for a historical time period. This project builds upon current projects focused on understanding 1) boreal aquatic ecosystem vulnerability to wildland fire and climate change, 2) freshwater habitat potential for Chinook salmon, and 3) drivers of Chinook Salmon productivity. Predictions will be applied across synthetic river networks developed from five-meter interferometric synthetic aperture radar (IFSAR) data for the YKRB. Important project outputs will include geospatial layers and an online results viewer of aquatic habitat potential for important commercial, sport, and subsistence fish species. Managers and researchers can use the project outputs to identify areas in the YKRB to focus future resources and effort for conservation and management of aquatic habitats and species.

Evaluating Environmental DNA as a Complementary Tool for Estimating Salmon Abundance in the Yukon River Basin

Student Investigator: Maggie Harings, MS Fisheries

PI: Andrés López, Erik Schoen

Funding Agency: USGS AK CASC (RWO 249), NSF GRFP, NIWR WERC

In-Kind Support: ADF&G, TCC, USFWS, BLaST, UAF URSA



Maggie Harings filters river water in the Chena River to collect salmon DNA in the system. Photo credit: Erik Schoen.

Across the Yukon River Basin, state, federal, and tribal entities estimate run timing and abundance of Chum (*Oncorhynchus keta*) and Chinook Salmon (*O. tshawytscha*) to inform in-season fishery management. Recently, more frequent high streamflow events and budget cuts have caused challenges for salmon escapement assessment projects, creating gaps in long-term datasets. To complement existing methods, we tested whether salmon abundance could be estimated from species-specific eDNA concentrations in river water. In 2021–2022, state, federal, and tribal partners

collected 728 water filtrate samples over 238 sampling days from five Yukon River tributaries. Chinook and Chum Salmon DNA was quantified via species-specific qPCR (quantitative polymerase chain reaction) assays. Field crews successfully collected clean eDNA samples, with only a 1.5% contamination rate among negative controls. Exploratory analyses showed a delayed increase in eDNA concentrations after spikes in daily salmon passage, consistent with prior studies in Alaska. Species- and river-specific generalized linear models were fit to test how closely eDNA, water temperature, and discharge predicted daily fish counts at weirs and counting towers on three tributaries. The most parsimonious daily model was for Chum Salmon counts in Henshaw Creek, where spawning occurs near the weir and eDNA sampling site. However, daily models generally had low predictive power for the other species and sites. We are now fitting time-series models to account for time lags between fish passage and eDNA detection to account for DNA released during spawning and carcass decomposition. Additionally, we are comparing eDNA concentrations before, during, and after the Chinook Salmon run in the Chena River (2022) and using ANOVAs to assess temporal differences in eDNA concentrations across lateral river transects. Long-term, we aim to build capacity for cost-effective salmon population monitoring to enhance salmon assessment capacity in Alaska's fisheries.

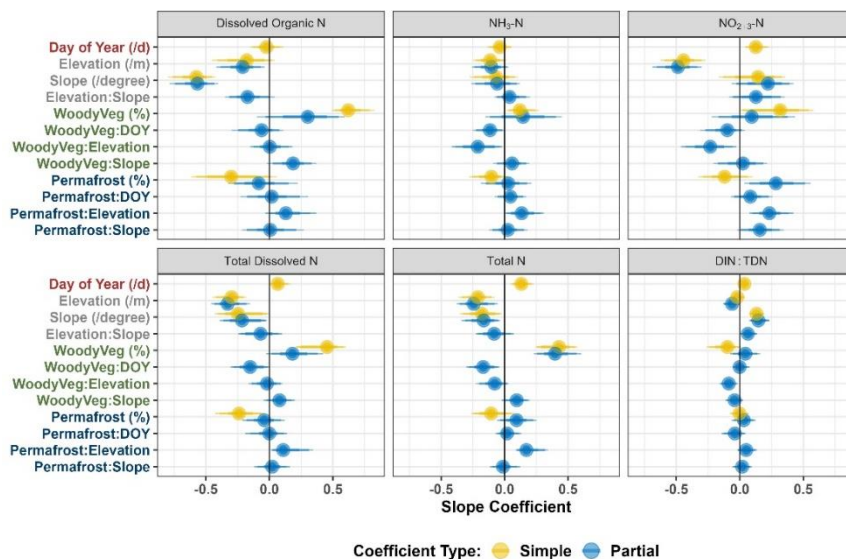
Spatial and Decadal-Scale Temporal Changes in Water Chemistry and Macroinvertebrates in Central Alaska

Collaborator: Caleb Robbins, Baylor University

PI: Jeff Muehlbauer

Funding Agency: NPS (RWO 254), BLM (RWO 258)

In-Kind Support: NPS



Modeled effects of day of year, elevation, woody vegetation (shrub and forest), and permafrost on concentrations of various nitrogen forms.

Northern latitudes are disproportionately warming due to climate change, with potential consequences cascading to aquatic life. To understand and manage climate-driven changes to its resources, staff led by Trey Simmons at NPS have been collecting water chemistry (carbon, nitrogen, phosphorus, cations, pH, etc.) and benthic macroinvertebrates from ~225 streams in the Central Alaska Inventory and Monitoring Network (parks and preserves: Denali, Wrangell-St. Elias, and Yukon-Charley Rivers) since 2004.

Investigator Robbins explored catchment-scale spatial drivers and decadal-scale temporal trends in the water chemistry and invertebrate communities. Modeling involved generalized linear mixed models fit in a Bayesian framework to account for site-level replication, non-Gaussian response families, censoring (water chemistry), and measurement error (invertebrate richness). Carbon and nitrogen analyte concentrations typically decreased with increasing elevation, and phosphorus and nitrogen have tended to increase through time. Woody vegetation and permafrost also had distinct controls on various forms of N, and nitrate concentrations generally increased from late spring to early fall. The most striking and consistent (across temporally replicated sites) changes were specifically in total dissolved nitrogen, often from near detection limits (<0.005 mg N/L) to 0.2 - 0.4 mg N/L within two decades. Elevation was also a dominant catchment-scale spatial correlate of invertebrate diversity. Richness increases at individual sites ranged from 0 to a doubling of taxa (e.g., from ~20-40) within two decades. There are likely multiple mechanisms driving benthic community change across the Central Alaska Network; in some sites, there are no changes in diversity, but there are significant, consistent changes in community structure (dominance). These analyses establish baselines for monitoring in a rapidly changing Alaskan climate, as well as inform research priorities for understanding and forecasting future change.

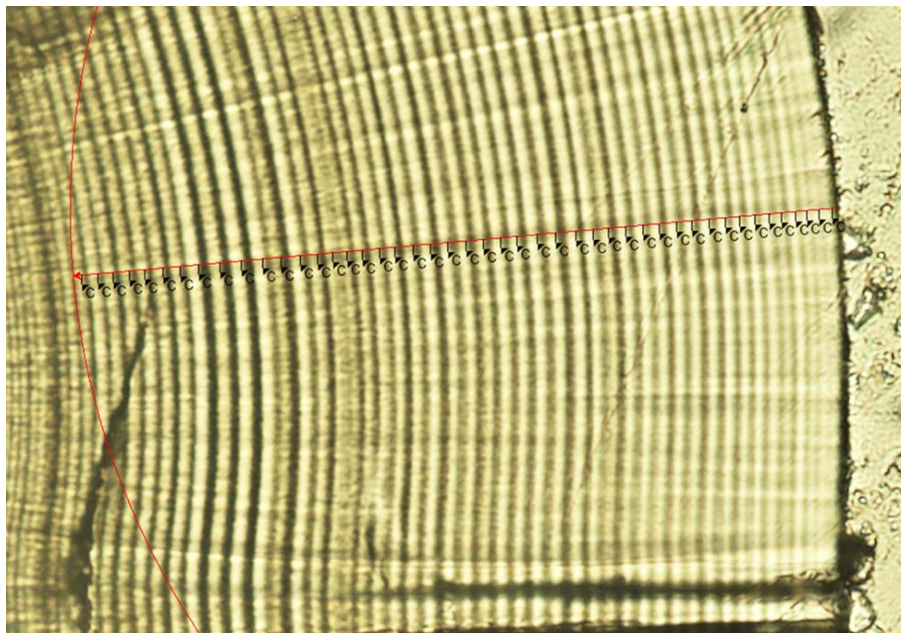
The Effects of Water Temperature and Heat Stress on Juvenile Chinook and Coho Salmon Growth in the Deshka River Watershed.

Student investigator: John Hermus, MS Fisheries

PI: Jeff Muehlbauer

Funding: USGS AK CASC (RWO 256)

In-Kind Support: USFWS



Juvenile Coho Salmon otolith with a transect line displaying the marked daily growth increments. Photo credit: John Hermus.

Warming climate and several declining salmon returns across the state of Alaska have led to an increased concern about the future of Alaskan salmon populations. This study considers sublethal effects of warm freshwater temperatures during juvenile rearing of two co-occurring Pacific Salmon species by linking water temperatures to growth rates. Juvenile growth of Pacific

Salmon plays a key role in their success because individuals that enter the ocean at larger sizes have higher survival rates. We hypothesized that Alaska's warmest freshwater temperatures already impair the growth of juvenile Pacific salmon due to direct effects of heat stress or indirectly via increased metabolic rates that exceed consumption. We sampled juvenile Chinook and Coho Salmon in the Deshka River, one of the warmest rivers in Alaska, where summer water temperatures routinely exceed 20°C. During the summers of 2019 to 2022 juvenile salmon were trapped at pre-established temperature monitoring stations throughout the Deshka River watershed. Of these fish, a subset was lethally sampled to obtain otoliths for daily growth analysis, stomachs for fullness, and muscle tissue for heat stress determination. These data provide a natural experimental window into juvenile salmon growth in some of Alaska's warmest waters. Such information will allow us to identify key water temperature thresholds associated with low habitat suitability to help inform future management and conservation efforts in a warming world.

Assessing Macroinvertebrate Community Response Following Stream Restoration to Guide Freshwater Conservation

Student Investigator: Dakota Keller, MS Biological Sciences

PI: Jeff Muehlbauer

Funding Agency: BLM (RWO 258), IAB Summer Research Fellowship, Golden Valley Electric Association, Ascentra Credit Union, DBW Nicholas Hughes Scholarship, URSA Mentorship Award, Calvin J. Lensink Endowment

In-Kind Support: USFWS, BLM, TVWA, IALT, AFS Hutton Scholars



A caddisfly under the microscope from one of the Cripple Creek macroinvertebrate samples. Photo credit: Dakota Keller.

Streams in Interior Alaska are of special focus for restoration to support fisheries resilience in the Yukon River, but are challenging to restore due to historic and ongoing placer mining and complex hydrology in the subarctic. Cripple Creek in Fairbanks, Alaska and Nome Creek in the White Mountain Recreation Area, 60 miles north of Fairbanks, are two restoration projects being studied for differences between their unrestored and restored reaches, using macroinvertebrate communities as a biological indicator of physicochemical habitat health and food web availability for fish. While it is expected that macroinvertebrate communities will rebound following stream restoration implementation it is unknown how long this will take following restoration disturbance, especially in Interior Alaska that has a limited growing season. Both case studies will assist restoration practitioners in accounting for potential delays in the biological response to restoration by assessing how long it takes macroinvertebrate

communities to rebound not just from degradation caused by mining, but also by disturbance from restoration activities. To measure this, macroinvertebrates were collected from restored and unrestored reaches in Cripple Creek, as well as canopy, turbidity, water temperature, and dissolved oxygen in both 2024 and 2025. Nome Creek was sampled similarly between its four total reaches. Macroinvertebrates from both sampling years are currently being processed in the lab and identified to genus for Ephemeroptera, Plecoptera, and Trichoptera and to family for other orders.

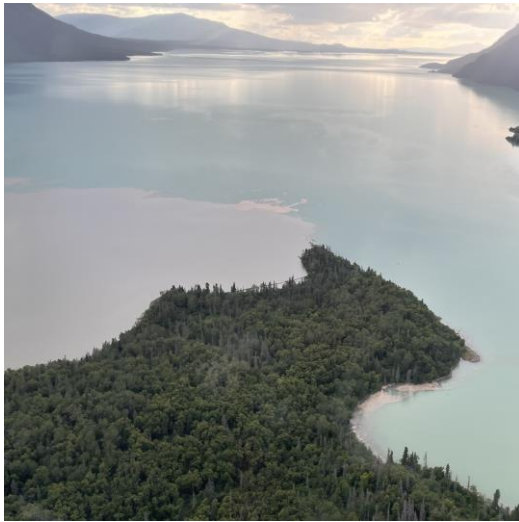
Climate and Landscape Influences on Lake Vertical Structure in Southwest Alaskan National Parks

Post-Doctoral Researcher: Matt Dunkle, IAB

PI: Jeff Muehlbauer

Funding Agency: NPS (RWO 259)

In-Kind Support: NPS



Glacial meltwater flowing into the Iliuk Arm of Naknek Lake. Photo credit: Matt Dunkle.

Climate change is reshaping the physical and chemical structure of subarctic freshwater ecosystems, with significant consequences for aquatic species such as Pacific Salmon. In southwest Alaska, lakes serve as integrators of watershed processes and sentinels of environmental change—particularly in terms of temperature, chemistry, and light availability. This project leverages a long-term, spatially extensive dataset of vertical water quality profiles collected through the National Park Service Inventory & Monitoring program, in collaboration with Krista Bartz, Sadie Textor, Paul Gabriel (Alaska Dept. Environmental Quality), Peyton Thomas (University of Colorado Boulder), and Sarah Laske (USGS Alaska).

Since 2008, four accessible lakes have been sampled annually, while 33 smaller, more remote lakes across

Lake Clark and Katmai National Parks have been sampled approximately once every three years. In the two largest systems—Lake Clark and Naknek Lake—profiles have been collected at 10 locations across multiple basins using a YSI ExoSonde 2 multi-sensor sonde. Measurements include temperature, dissolved oxygen, pH, conductivity, turbidity, and ancillary conditions such as wind speed and Secchi depth.

The goal of this work is to assess long-term changes in water quality and vertical structure, with an emphasis on habitat suitability for juvenile Sockeye Salmon and other fish species. We are currently validating semi-mechanistic models of lake temperature structure using downscaled climate and morphometric data. Preliminary results show increasing surface temperatures and altered thermal stratification patterns, especially during warm PDO phases. Future work will apply modern statistical approaches, including sparse regression, to identify key drivers of observed changes. This research will help anticipate ecological tipping points and improve understanding of how climate, landscape features, and mountain water sources interact to shape subarctic lake ecosystems. This work has resulted in an accepted article in the Alaska Parks Science Journal.

Rusting Rivers in Northwest Alaska Impacts on Stream Biota

Student Investigator: Lauren Yancy, MS Fisheries

PI: Jeff Muehlbauer

Funding Agency: USGS AK CASC (RWO 262)

In-Kind Support: ADF&G



Rusting river in the Wulik River drainage. Photo credit: Lauren Yancy.

The Arctic is experiencing rapid warming, which is altering seasonal weather and, in turn, influencing hydrologic processes. In the Alaskan Arctic, continuous permafrost is thawing and expanding the Earth's active layer, creating new hydrologic connections that are mobilizing previously sequestered heavy metals. The resulting heavy metal seeps flow into previously pristine Arctic streams, creating a visually striking phenomenon we call "rusting rivers" due to the high iron content. Over the past decade, rusting rivers have been increasing in frequency, raising concern for the health of the stream ecosystem,

many of which support subsistence and sport fisheries. This project leverages an existing biomonitoring program conducted by ADF&G in the Wulik River drainage, while introducing a new component focused on assessing the health of juvenile Dolly Varden in rusting rivers. The project goals are to (1) characterize spatial and temporal patterns in biological metrics in relation to existing and newly observed rusting seeps, and (2) document histopathological impacts of rusting rivers on juvenile Dolly Varden. Preliminary findings suggest that streams affected by any level of rusting experience declines across all biological metrics (i.e., periphyton density, macroinvertebrate abundance and richness, and fish abundance). In 2025, juvenile Dolly Varden gill and liver samples were collected from fish captured in waters impacted by rusting seeps and will be analyzed for histopathological changes.

Linking Macroinvertebrate Communities and Fish Bioenergetics in a Mined Subarctic Stream

Student Investigator: Thomas House, MS Fisheries

PI: Jeffrey Muehlbauer

Funding Agency: NPS (RWO 263)

In-Kind Support: NPS



Drift nets deployed in Yukon-Charley Rivers National Preserve. Photo credit: Thomas House.

Coal Creek is a former placer mine located on the upper Yukon River, roughly 60 miles upstream from Circle, Alaska. In addition to its role as an important cultural and historical site, Coal Creek represents potential rearing habitat for juvenile Chinook salmon (*Oncorhynchus tshawytscha*), a species that has undergone dramatic declines across the Yukon River basin. However, decades of placer mining have altered stream morphology in ways that likely affect the macroinvertebrate food base needed to support juvenile growth.

Preliminary sampling has already shown higher abundance and diversity of macroinvertebrates in nearby unmined reference streams, suggesting long-lasting impacts from mining. To better understand these long-term effects (50+ years post-mining) on rearing fish populations and their prey base, we launched a study to model juvenile salmonid growth potential in Coal Creek. We are doing this by quantifying both abundance and energy density of macroinvertebrate prey to parameterize a fish

bioenergetics model.

This summer, we completed field sampling at Coal Creek as well as two unmined reference streams: Sam Creek and Thanksgiving Creek. We measured macroinvertebrate prey availability using drift and Surber samples, collected fish diet samples via gastric lavage to validate prey selection, and recorded water quality and riparian cover in each reach. With fieldwork successfully wrapped up, the collected samples will now be processed and analyzed this fall, laying the groundwork for a clearer understanding of how legacy mining continues to shape stream ecosystems and salmon production potential in the Yukon River basin.

Distributions and Ecology of Arctic and Pacific Lampreys in Alaska

Student Investigator: Nate Cathcart, PhD Fisheries

PI: Trent Sutton, CFOS and Jeff Muehlbauer

Funding Agency: BIA

In-Kind Support: ADF&G, Knik Tribe, CRITFC



A Pacific Lamprey on its nest in an Alaskan river. Photo credit: Nate Cathcart.

Arctic Lamprey (*Lethenteron camtschaticum*) and Pacific Lamprey (*Entosphenus tridentatus*) in Alaska arguably represent the state's most understudied yet widespread anadromous fishes. Although their ranges overlap with Pacific salmon, the distributions and ecologies of these species remain poorly understood. Without better knowledge of basic natural history, Alaska's resource managers will not be able to mitigate potential effects of local environmental differences and development on lamprey populations. The primary goals of this project are to improve maps of lamprey

distributions and determine life-stage-specific ecological patterns. This will be done by exploring larvae (e.g., growth, genetic pedigrees), juveniles (e.g., timing of transformation, migration), and adults (e.g., spawning migration patterns, genetic baselines/pedigrees, and spawning site selection). In addition to Knik Tribe, ADF&G, and CRITFC, this project includes collaboration with many other tribal, federal, private and local stewards, and integrates local knowledge, electrofishing, fish wheels, observations, sonar footage analysis, and genetic analysis. Although the work is statewide, the focus is on southcentral Alaska in watersheds of Cook Inlet and the Copper River basin. Thus far, larvae have been collected frequently across our study area for both Arctic and Pacific lampreys. Adult lamprey spawning and migrations have been best documented for Pacific Lamprey. To date, juveniles have only been collected for Arctic Lamprey. Future sampling will continue our vision toward providing a clear roadmap for future conservation, research, and management of Alaska's lampreys.

Completed Wildlife Studies

Investigating the Role of Juvenile Weight in Predicting Future Fitness in Northwestern Alaska Moose Populations

Student Investigator: Sara Henslee (Germain), MS Wildlife Biology and Conservation

PI: Shawn Crimmins

Funding Agency: ADF&G (Wildlife Base)

In-Kind Support: ADF&G Region V



Radio-collared moose calf near Nome, AK. Photo credit: S. Henslee.

Moose are a valuable big game species in Alaska and serve as a critical food resource for rural communities. Moose densities near Nome, Alaska have remained low for decades despite no indications of nutritional limitations to fitness. Understanding the limitations on moose population growth is critical for developing appropriate habitat and harvest management strategies aimed at increasing harvest opportunities for local residents. Current metrics of nutritional limitations do not explain the stable low-density seen among moose populations near Nome, requiring a better understanding of the links between physical condition and demographic rates. The objectives of this study are to 1) quantify neonate moose survival in Game Management Unit 22, and 2) identify factors associated with neonatal survival, and 3) relate physical condition metrics to long-term reproductive output of adult moose.

We captured and monitored neonate moose in Game Management Unit 22 and monitored their survival through recruitment the following year. We used Cox Proportional Hazards models to quantify the effects of maternal body condition and habitat on neonate survival. We used long-term monitoring data to evaluate the influence of calf body condition on early reproductive success using logistic regression models. Our results suggested a positive relationship between short-yearling weight and age to primiparity (i.e., first calving event) and twinning rates through at least age 7. However, we noted interannual heterogeneity between the probability of being primiparous at 2- and 3-years-old, suggesting primiparity may be influenced by environmental variables not investigated in this study. Results of moose calf survival suggest that in a predator-limited moose population, survival is random and independent of maternal characteristics. This study will help identify the causes behind low, long-term moose densities in Game Management Unit 22, and also help to refine nutritional and body-condition metrics used to assess nutritional limitation for moose in Alaska. Identification of the reasoning behind low moose densities will help guide management actions aimed at increasing densities and, thus, harvest opportunities.

Movement and Habitat Use of Female Sea Otters (*Enhydra lutris*) In Kachemak Bay, Alaska

Student Investigator: Luke Porter, MS Wildlife Biology and Conservation

PI: Greg Breed, DBW, IAB

Funding Agency: USFWS (RWO 247)

Understanding how, when and why animals select and use various habitats is crucial in ecology, influencing individual fitness, population dynamics, and informing conservation and management strategies. This is especially important for species like sea otters (*Enhydra lutris*), which play a significant role in their ecosystems. Habitat preferences often vary throughout any year or an animal's life stages due to differences in resource needs, predation risks, and competitive interactions. These variations can lead to individual specialization and heterogeneity in habitat use, diets, and life history strategies, affecting the broader ecological community and trophic interactions.

Reproductive strategies and success significantly influence population dynamics. Females are critical to population health, particularly in species with polygynous mating systems, where males do not contribute to parental care. Consequently, changes in female mortality or habitat quality can have profound long-term effects on population stability. For species that carry their young during the dependent period, females must navigate the dual challenges of securing resources and minimizing predation risks for both themselves and their vulnerable offspring.

Sea otters are uniquely sensitive to habitat quality due to their high metabolic rates and the lack of blubber, relying on dense fur for insulation. As extreme income-breeders, they must forage constantly to meet the energetic demands of lactation. Female otters with dependent pups face even greater challenges, requiring habitats that provide ample resources while offering protection from predation and environmental stressors. Understanding the specific habitat needs of female otters with pups is vital to their conservation, as disruptions in their habitats can increase mortality risks and affect population growth. Sea otters are also keystone species and ecosystem engineers, exerting considerable top-down control in coastal ecosystems by preying on sea urchins, which in turn allows the growth of kelp forests. These ecological interactions underscore the importance of understanding habitat use patterns in different life stages and demographic segments.

Habitat selection is influenced by several factors, including age, sex, population density, and reproductive status, leading to complex spatial and temporal patterns. Seasonal changes in environmental conditions further complicate habitat dynamics. These seasonal density fluctuations are critical to understanding population health and the ecological processes that sustain it.

In Kachemak Bay, Alaska, sea otters have reestablished significant populations, presenting an opportunity to study their habitat use and movement patterns in a region characterized by complex coastlines. By examining seasonal variations in otter habitat and space use, we can better understand how environmental changes impact otter populations and their ecological roles. This knowledge is crucial for informing conservation strategies, ensuring that key habitats are protected to support the health and stability of otter populations and the broader ecosystem they help sustain.

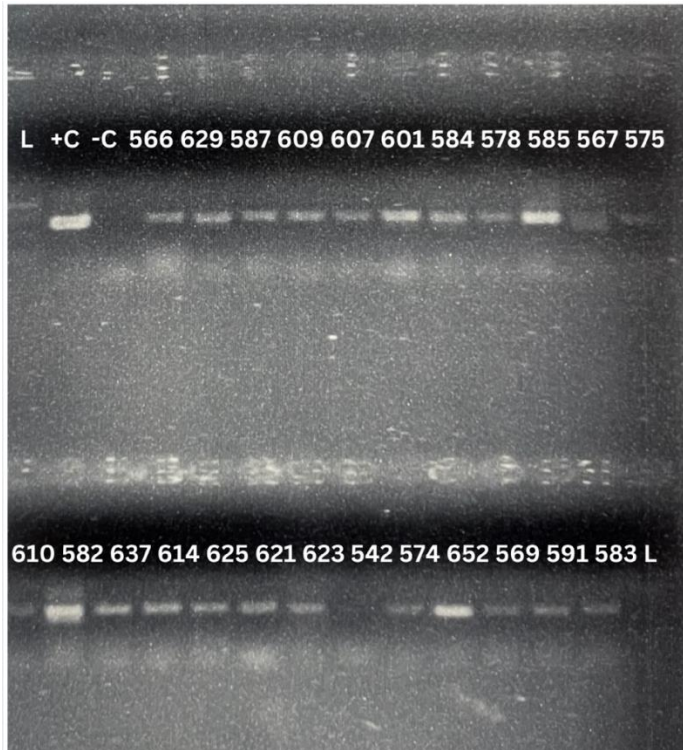
Small Mammal Disease Prevalence in Interior Alaska

Student Investigator: Tabitha Page, M.S. Wildlife Biology and Conservation

PI: Shawn Crimmins and Nina Hansen, VetMed

Funding Agency: UAF VetMed

In-Kind Support: UAF VetMed, UAF DBW



Amplification of fragment of ITS region showing positive amplification for bartonella. Photo credit: Tabitha Page.

Climate change is disproportionately affecting the Arctic and subarctic. In addition to altering the distributions of flora and fauna, a warmer climate threatens to expand the geographic range of a number of zoonotic diseases (diseases transmitted to humans from animals). In particular, zoonotic diseases that can survive in the environment (i.e. surface water), or in arthropod vectors (ticks, etc.) are of concern. Two diseases of particular concern in Alaska are the bacterial zoonoses leptospirosis and bartonellosis. Although these diseases are known to occur in wildlife in interior Alaska, very little is known of their distribution, prevalence, or risk factors.

We collected small mammals from three areas with differing levels of potential risk factors associated with disease transmission: sled dog kennels, captive animal facilities, and reference sites.

Once captured and euthanized, specimens were dissected and organ tissue (liver, kidney, spleen) was used for DNA extraction using PCR to amplify *Leptospira* or *Bartonella* DNA, with any positive samples being sequenced to identify specific bacterial species and serovar.

We found no evidence of leptospirosis but did find that bartonella prevalence varied by site type and was positively associated with the presence of ectoparasites (fleas). Our results suggest that parasites may serve as reservoirs and/or vectors for *Bartonella* in Alaska small mammals.

Ongoing Wildlife Studies

Evaluating Spatial-Capture-Recapture Models for Social Carnivores

Post-Doctoral Researcher: Derek Arnold

PI: Shawn Crimmins

Funding Agency: ADF&G

In-Kind Support: ADF&G



Group of wolves at rendezvous site on Prince of Wales Island.
Photo credit: M. Kampnich.

Spatially-explicit-capture-recapture models are a powerful tool for estimating population density and for understanding the drivers of variation in density. These tools expand on classical mark-recapture methods by incorporating information about animal movements and space-use into the estimation process, providing more precise and less-biased estimates of density. However, violations of sampling assumptions can lead to erroneous conclusions regarding population

density and its drivers. The results of this project have direct applicability to wildlife monitoring not only in Alaska, and not only for wolves, but for a broad range of species and systems. The sampling issues present for wolves in Alaska are not unique, and a robust understanding of how these violations affect modeling results when using SECR has the potential for improving wildlife monitoring globally.

The use of SECR models in Alaska is increasing, with recent applications to wolves in many different game management units. Recent concerns over the accuracy of density estimates derived from SECR models for wolves in Alaska have led to a concern that sampling violations may be occurring when using hair boards to sample these socially structured and highly mobile animals. Using a combination of simulations and empirical data from multiple study sites throughout Alaska, this project seeks to develop new statistical models to determine the degree to which social structure (i.e. subordinate individuals not being sampled) and unique movement behaviors influence SECR estimates.

Preliminary results from these new models suggest that movement behavior and social status can affect detection rates, and that this variation can be accounted for using a newly developed hierarchical SECR model.

Big Game Movement and Vital Rate Modeling

Post-Doctoral Researchers: Alison Ketz, Derek Arnold

PI: Shawn Crimmins

Funding Agency: ADF&G

In-Kind Support: ADF&G



Moose (*Alces alces*) in interior Alaska. Photo credit: Shawn Crimmins.

In recent years ADF&G completed a large-scale moose radio-collaring project in the Alphet Hills area of southcentral Alaska. This project represents a substantial source of spatial and demographic information for moose in this region, which is of particular interest due to previous habitat management actions for improving moose forage. In collaboration with ADF&G, we will develop habitat, movement, and survival models for moose from this existing dataset in order to better understand the demography and habitat associations of moose in this region.

We used a Bayesian time-to-event framework to assess how environmental, demographic, and behavioral factors affected survival. Preliminary results suggest that moose which displayed migratory behavior and used a prescribed burn area had lower hazard (i.e. higher survival). Moreover, female moose had a lower baseline hazard than did male moose, and higher hazards were associated with higher levels of precipitation, potentially indicating a negative effect of rain-on-snow events. A manuscript related to this survival modeling is currently in development.

We are currently in the process of developing integrated step-selection functions for moose from this dataset in order to determine how movement rates and general habitat use patterns are influenced by migratory behavior and how moose use the prescribed burn area. A manuscript related to this analysis will be developed and submitted within the next year.

Factors Limiting Population Growth of Unhunted Dall Sheep

Student Investigator: Christie Osburn, MS Wildlife Biology and Conservation

PI: Shawn Crimmins and Megan Milligan

Funding Agency: ADF&G (Wildlife Base)

In-Kind Support: ADF&G



Dall sheep ewe foraging. Photo credit: ADF&G.

Despite nearly a decade free of harvest pressure, sheep populations in northwest Alaska have not rebounded as anticipated. Identifying the factors that continue to repress these populations will aid in the general understanding of Dall sheep ecology and help inform how sheep populations, at the margins of their range, can be sustainably managed in the future. Determining the relative effects of these factors can assist managers in determining appropriate harvest seasons and bag limits and identifying critical landscape features and areas requisite to the

continued presence of sheep in GMUs 23 and 26. The objective of this study is to model the habitat selection, survival patterns, demographics, and nutritional health of the De Long and Baird Dall sheep herds. We will radio-collar approximately 20 sheep in each of two herds (De Long, Baird) in the fall/winter of 2023 and 2024. Mortality will be monitored via movement data and mortality sites will be visited as soon as possible to attempt to determine mortality cause. If possible, tissue samples will be collected for additional analysis. Health assessment will be conducted during capture and biological samples collected to analyze for disease, genetics, pregnancy, and nutritional limitation. Collared ewes will be radiotracked with fixed-wing aircraft weekly between May and June to determine lambing rates using the presence of a lamb at heel.

Capture efforts in 2023–2025 were unsuccessful due to a combination of poor weather conditions and pilot availability. This study will be the first to quantify habitat use patterns, survival rates, and health of sheep in these herds. This project will provide ADF&G with the information needed in order to manage both of these sheep herds in order to improve population trajectory.

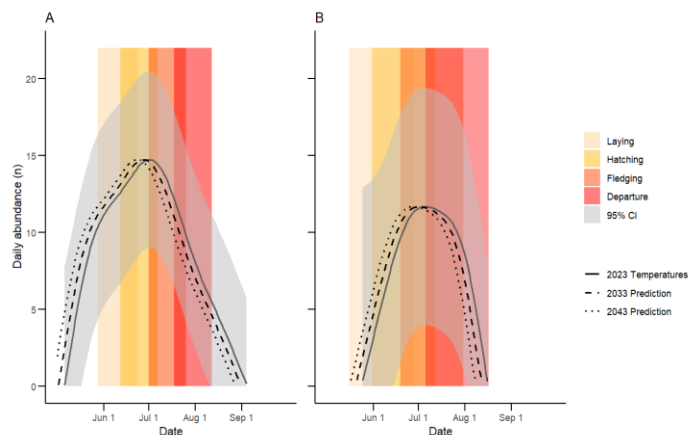
Investigating Aerial Insectivore Phenology as Related to Insect Abundance and Richness Within Boreal Alaska

Student Investigator: Ian Kwit, MS Biological Sciences

PI: Jeff Muehlbauer

Funding Agency: ADF&G (Wildlife Base), NSF EPSCoR, IAB

In-Kind Support: INBRE, ADF&G, VetMed, DBW



Modeled daily emergence compared with regional *C. cooperi* breeding schedules in Anchorage (A) and Fairbanks (B).

Migratory aerial insectivorous birds serve as a critical link between river ecosystems and their neighboring terrestrial habitat, facilitating nutrient transport by foraging on emerging aquatic insects. Within the last several decades, insectivorous bird species across the United States have experienced substantial population declines, potentially jeopardizing the stability of such aquatic ecosystems. As insect emergence is a strongly temperature dependent phenological event, the warming of ambient

temperatures in northern latitudes is not without consequence for rivers and the insects who inhabit them. With warming ambient temperatures pushing insect emergence earlier into the calendar year, there is concern that migratory insectivorous birds are arriving out of sync with prey they are used to foraging on, depriving them of key nutrients, lowering health of adults and juveniles, and driving observed population declines. In response, ADF&G launched a study to investigate the availability of insect prey around nesting sites of one such migratory insectivore, the Olive Sided Flycatcher (*Contopus cooperi*) during their breeding season. Data on the phenological timing of the birds' breeding season was collected at known nests (84 nests total: Anchorage n = 22, Fairbanks n = 40, Tetlin n = 22). The ambient insect community surrounding these known nests and sites historically used (108 sites total: Anchorage n = 17, Fairbanks n = 74, Tetlin n = 17) were sampled using hanging Malaise and canopy traps during the summers of 2013-2018, offering a direct insight into the insect community available for foraging and breeding *C. cooperi*. This data was paired with aquatic insect samples collected from emergence traps deployed on the waters of Chester, Campbell, and Caribou-Poker Creeks in Fairbanks and Anchorage during the summers of 2023 and 23. These emergence traps offer information on how aquatic insect emergence patterns change over the summer and as stream temperatures warm. Using data on aquatic insect emergence gathered using these traps, estimates of the availability of aquatic insects to foraging *C. cooperi* were constructed and plotted against known timing of breeding stages, shown in the figure above. These models offer insight into the level of asynchrony and let us consider implications for adult and chick health if aquatic insect emergence continues to be pushed earlier into the year. Warming's effects on this relationship were simulated by adjusting regional Cumulative Degree Day schedules.

Synthesizing Ungulate Movement and Migration Across Alaska

Student Investigator: Miles Spathelf, PhD Biological Sciences

PI: Shawn Crimmins and Megan Milligan

Funding Agency: ADF&G

In-Kind Support: ADF&G



Caribou migrating during early spring. Photo credit: Shawn Crimmins.

Ungulates, especially caribou, in Alaska have some of the longest terrestrial migrations in the world. Human perturbations from roads, mining efforts, hunting, and oil development all have an influence on these vast movements on the landscape undergoing climate change. GPS telemetry data has been collected for many years on numerous herds and has yet to be analyzed over the long-time scale ADF&G's data covers. Both movement corridor and barrier behavior analyses can shed light on impacts ungulates are facing

when moving between areas critical for their life histories. Appropriate management of harvested species requires understanding of where these creatures migrate and what barriers they encounter. Vast stores of energy are used for these movement behaviors and can drastically impact survival and reproduction. The objective of this study is to analyze centuries of caribou-years of GPS telemetry data for changes in migration corridors and barrier behaviors as they have arisen on the landscape. Telemetry data from multiple studies and biologists will be compiled and organized for movement analyses and behavior analyses providing both analytical output as well as a dataset for future studies to be conducted more readily. We have obtained, organized, and cleaned decades of caribou herd location information and are conducting movement corridor analyses and will be completing adapted barrier behavior analyses. The results of this project will provide corridor information to inform future infrastructure work to help reduce impacts on highly variable herds that hunters rely on for subsistence or are supplemental to hunter's food security. The results may also be useful in comparing ungulate crossing strategies that lessen impact on ungulates.

Developing a Multi-Species Monitoring Framework in Southeast Alaska

Student Investigator: Sarah Trujillo, PhD Biological Sciences

PI: Shawn Crimmins

Funding Agency: ADF&G (RSA S298086, RSA S29254)

In-Kind Support: ADF&G



Wolves captured on a remote camera on Prince of Wales Island.

Photo credit: Gretchen Roffler, ADF&G.

Effective monitoring of large mammals in southeast Alaska is challenging due to accessibility and heavy forest cover. Typical monitoring approaches for wolves, black bear, and deer are largely ineffective in this region, making effective management actions challenging. Moreover, recent listing petitions for wolves in this region necessitate the development of an effective approach for identifying population status. Appropriate

management of harvested species necessitates a robust understanding of population status and change through time. In the absence of such information, management programs are hindered in their ability to appropriately allocate harvest or to assess the vulnerability of threatened populations. Thus, a robust monitoring framework for large mammals in southeast Alaska is critical not only for harvest management, but also for status assessments of potentially threatened species. The objective of this study is to develop an integrated monitoring framework for large mammals on Prince of Wales Island, with an emphasis on wolves, that can be applied throughout southeast Alaska. We are using a combination of motion-capture and time-lapse images to develop density models for wolves, black bear, and black-tailed deer using a suite of analytical approaches including recently developed space-to-event and time-to-event models. We were able to deploy 21 cameras throughout Prince of Wales Island during our pilot field season in fall of 2021, 50 cameras in summer of 2022, 83 cameras in summer of 2023, and 100 cameras in summer 2024. All photos from 2021–2023 have been tagged and processed. The results of this project will provide the basis for effective long-term monitoring of ecologically, economically, and culturally important large mammals in southeast Alaska. Additionally, this project will provide information on the ecological relationships between predator and prey species in southeast Alaska.

Black-Tailed Deer Density, Habitat Use, and Predator-Prey Dynamics

Student Investigator: Tony Zambito MS Wildlife Biology and Conservation

PI: Shawn Crimmins

Funding Agency: ADF&G

In-Kind Support: ADF&G, UAF DBW



Sitka black-tailed deer captured on a remote camera on Prince of Wales Island. Photo credit: Shawn Crimmins.

Effectively monitoring Sitka black-tailed deer in southeast Alaska is difficult due to inaccessibility and heavy forest cover, making effective management actions challenging. Recent listing petitions for wolves in this region also necessitate an effective approach to monitoring deer, which are an important prey species for wolves. Deer are an important game species in southeast Alaska, and a critical driver of predator population dynamics in the region. On Prince of Wales Island (POW), where habitat management for deer is a major

land use issue, it is critical to understand where deer occur on the landscape and how they respond to timber harvest. Our understanding of how predators shape deer distribution on the landscape could also have important implications for harvest and habitat management. The objective of this study is to build on previous research that has demonstrated that deer respond to habitat management in terms of browsing pressure by also linking overall occupancy rates and interactions with predators to habitat conditions. We are using motion-capture and time-lapse images from cameras deployed in summer of 2023 to develop single-season and multi-species occupancy models for deer. These models will quantify occupancy rates across the landscape and how these rates are affected by habitat conditions and the presence of predators. Preliminary results suggest that deer habitat use is affected by the presence of predators but that this is mediated by habitat conditions and varies seasonally. The potential for non-invasive multi-species monitoring frameworks could provide a cost-effective mechanism for assisting with harvest and/or habitat management decisions. Additionally, this project will provide information on the ecological relationships between predator species and deer in southeast Alaska.

Human-Bear Conflict in Juneau, AK

Student Investigator: Binta Wold, MS Wildlife Biology and Conservation

PI: Todd Brinkman, IAB/DBW

Funding Agency: ADF&G



Survey participation sticker. Design credit: Binta Wold.

This project is focused on advancing knowledge on human-bear conflict in Juneau Alaska. Our objectives were to quantify perceptions on: the extent of conflict, factors contributing to conflict, and potential solutions. During spring of 2024, we conducted an online survey that included 5,000 randomly selected Juneau residents.

Approximately 750 completed people completed the survey. Sixty-nine percent of participants agreed that there are problems between people and bears in Juneau. The biggest problem was bears getting into trash (90% reported moderate to very serious problem) and other household attractants, followed by people getting to close bears (68% reported moderate to very serious problem). Survey respondents were more concerned about conflict leading to bears being killed (79%) than concerned about the safety of

people around bears (53%). Most respondents were satisfied (57% satisfied vs. 12% unsatisfied) with how ADFG manages bear in Juneau. Of 11 different interest groups relevant to human-bear conflict in Juneau, respondents most agreed that individual Juneau residents (84% agreed) and landlords (82% agreed) need to do more to fix the problem.

Small Mammal Population Dynamics and Genetic Variation Prior to Development in Alaska National Parks

Student Investigator: Sarah Swanson, PhD Biological Sciences

PI: Shawn Crimmins and Greg Breed, DBW, IAB

Funding Agency: USGS (RWO 261)

In-Kind Support: NPS, UAF DBW



Small mammal captured in Denali, National Park. Photo credit: Mel Flamme, NPS.

Small mammals play a vital role in boreal and Arctic ecosystems in Alaska. As local and resident keystone species in food webs with quick generation times and small home ranges, small mammals are sensitive bioindicators of local disturbance, including climate change and development. They live year-round in soils and beneath snow, reproduce prolifically and thus, may reveal responses to local environmental changes more quickly than larger fauna. As primary consumers, they influence many trophic levels, including insects, mushrooms, plants, seeds, and bark. Small-mammal biota are vital prey for over-wintering species (raptors, meso-carnivores, carnivores). Despite their importance in the Arctic and boreal food webs, vital ecological data on small mammals' populations in Alaska remains scant.

In collaboration with the Nebraska Cooperative Fish and Wildlife Research Unit, we will use advanced genetic approaches to quantify genetic diversity indices, genetic structure, spatial genetic structure, and effective population size across time using archived samples collected as part of Denali National Park's long-term small mammal monitoring program. We will identify individuals with mixed ancestry (if present) and use simulated assignment probabilities for first generation hybrids and backcrosses to better determine whether interspecific admixture assignments are due to hybridization or retained ancestral variation.

The results of this study will provide some of the first information regarding genetic structure of cyclical small mammal populations in Alaska. This information is critical for identifying ecosystem status and reference conditions within Alaska National Parks in the event of serious disturbance or construction efforts, such as the recently proposed Ambler Road corridor project within Gates of the Arctic National Park.

Distribution and Habitat Use of Nesting Surfbirds in the Steese National Conservation Area

Student Investigator: Sam Simon, MS Wildlife Biology and Conservation

PI: Shawn Crimmins and Greg Breed

Funding Agency: BLM (RWO 264)

In-Kind Support: BLM, USFWS



Viewing a surfbird (*Aphriza virgata*) in the Steese NCA. Photo credit: Sam Simon.

As part of the BLM's Strategic Plan for Migratory Bird Conservation, an understanding of the distribution and status of alpine nesting shorebirds within the Steese National Conservation Area (NCA) and White Mountains National Recreation Area have been identified as critical management needs for the Eastern Interior Field office. Very little is known of the distribution or nesting habitat of alpine nesting shorebirds in the Steese NCA, or more broadly in interior Alaska.

This project will entail surveys of upland and alpine habitats focused within the

Steese NCA (where access can be obtained from trails and nearby roads) to document presence and absence of alpine shorebirds and the distribution of territories and nesting locations. We will also capture surfbirds during nesting and brood-rearing to collect breeding ecology and demographic data and will affix satellite transmitters to a subset of these captured individuals to quantify habitat use, movements and migratory pathways for these individuals. We will develop habitat models using data from satellite tagged individuals to determine summer nesting and brood rearing habitat associations. Habitat use information will be used to quantify the probability of use of surfbirds across the Steese and White Mountains. Preliminary habitat models indicate that nesting surfbirds are exclusively located in high-elevation alpine habitats associated with rocky outcroppings. The information gained from this project will help BLM achieve their management and conservation objectives while also filling the identified role of National Landscape Conservation System lands as outdoor laboratories. This information will also supplement data being gathered on these alpine shorebirds during migration and wintering, allowing a better assessment of the limitations and importance of the Steese NCA in maintaining stable population sizes.

Effects of Human Visitation on Bear Activity at Anan Observatory

Student Investigator: Kayleigh McCarthy, MS Wildlife Biology and Conservation

PI: Shawn Crimmins

Funding Agency: USFS

In-Kind Support: USFS, UAF DBW



Black bear feeding on salmon at the lower falls of Anan Observatory. Photo credit: Shawn Crimmins.

The impacts of recreational activity on wildlife can be substantial. In areas with high concentrations of humans and wildlife, there may be high potential for human activity to negatively affect wildlife. At the Anan Observatory, near Wrangell, Alaska, the USDA Forest Service recently renovated their viewing platform at a substantial cost to the ranger district. The Anan Observatory is a highly regulated bear viewing site in the Tongass National Forest, with visitation controlled through a permit system, however the construction of a larger viewing platform has led

to calls for increasing the number of permits, and thus the human visitation rates. We are using remote cameras deployed at the observatory and at the upper falls, an area off limits to recreational visitation approximately 0.5-km upstream, to quantify bear activity patterns during the tourist season. We will use a combination of generalized linear modeling and activity pattern analysis to determine if activity levels differ between the control area (upper falls) and the observatory, and if bear activity is related to hourly human visitation rates or varies between species. We will also use long-term monitoring data from the observatory to determine if chronic human visitation has affected bear feeding activity. Preliminary results suggest a positive relationship between human visitors and bears at the observatory, while there also appears to be a segregation between black back and brown bear use of areas that are not impacted by humans. The results of this study will provide the USDA Forest Service with important information that they can use to determine if increased permitting will have deleterious effects on bears at Anan Observatory. There is substantial financial pressure surrounding the management of the observatory, with a number of guiding services operating out of the town of Wrangell, and their financial security is contingent on a long-term management plan that maximizes the number of visitors without having any adverse effects on wildlife.

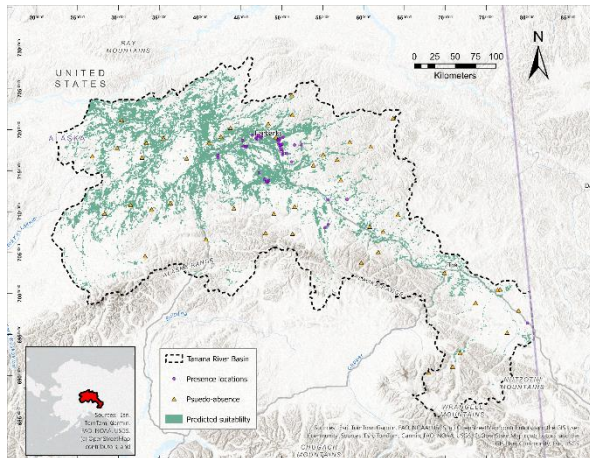
Little Brown Bat Habitat Association Modeling & Geodatabase Development

Investigators: Kara White (PhD), Matt Kynoch (MS)

PI: Shawn Crimmins

Funding Agency: USFWS

In-Kind Support: US Army Garrison Alaska – Fort Wainwright



Predicted suitable habitat for little brown bat (*Myotis lucifugus*) in the Tanana River Valley.

The U.S. Fish and Wildlife Service's Fairbanks Field Office, Ecological Services, proposes to assist the US Army Garrison Alaska (USAG Alaska) in refining their fish and wildlife database by identifying GAPs and collating data regarding fish, wildlife, and the habits they occupy. Accurately describing species' association with their habitat is integral to managing impacts on military lands. The USAG Alaska has invested resources to obtain land classification and habitat information, as well as bird and bat habitat use and procuring high quality imagery and LiDAR (light detection and ranging) data to assess changes to habitat over

time. Most studies are over 20 years old, and as a proactive measure, USAG Alaska requires habitat and association information updates and additions to revise current remediation and restoration methods to minimize impacts (Adaptive Management).

The proposed project will collate existing data and produce a geodatabase critical for evaluating future projects and to build little brown bat (*Myotis lucifugus*) habitat models in order to identify additional areas important to this species, which is currently under review for listing on the Endangered Species Act (ESA). The information will lead to increased mitigation success while adapting to a changing climate. Information will include ground truthing previous habitat assessments and reviewing current available LiDAR and imagery data. Information collation, geodatabase development, and map production will then assist in revising standard operating procedures (SOPs) and developing impact avoidance and minimization measures.

Preliminary results suggest that little brown bat occurrence in the Tanana River Valley is positively associated with deciduous trees and negatively associated with shrub cover. Future work will assess occupancy patterns of little brown bats and habitat associations of migratory birds.

Ongoing Ecological Studies

Arctic Biogeochemical Response to Permafrost Thaw (ABRUPT)

Research Technician: Colin Edgar, IAB

PI: Eugénie Euskirchen, IAB

Funding Agency: USGS (RWO 251), NSF LTREB, NSF LTER



The Bonanza Creek Fire approaching the field sites during June 2025. Image from the webcam installed at the internal collapse scar bog site.

Our objective is the continued collection of eddy covariance and associated meteorological and biophysical information across four sites spanning a gradient of permafrost regime to better understand the controls over carbon, water, and energy cycling in a lowland boreal landscape. Our sites include a black spruce forest with permafrost, two internal collapse scar bogs of different ages formed following thermokarst, and a rich fen without near-surface permafrost. These data have been collected beginning in 2010.

The Bonanza Creek Fire began on June 21, 2025 (shown above), with the fire coming within about one kilometer from our four sites. Access to the sites was restricted until mid-August 2025. Fortunately, our sites were not impacted due in large part to close coordination between Bonanza Creek LTER personnel and fire managers and the establishment of a containment line. Our instruments continued to run successfully, with sustained measurements of the carbon, water, and energy fluxes.

Our data have been made available through the Ameriflux website (<https://ameriflux.lbl.gov/>), including recent measurements collected through 2024. We are currently processing data from January 2025 – August 2025 in an effort to make data available for a 2025 synthesis of data across flux tower sites around the Arctic. In 2024, our data were included in the Arctic Report Card (<https://arctic.noaa.gov/report-card/report-card-2024/arctic-terrestrial-carbon-cycling/>), which showed that the Arctic has become a net source of carbon due to both wildfires and emissions of methane. These findings are also part of another publication that is currently in review at *Environmental Research Letters*. Furthermore, our flux data are included in a recent study of sublimation of tundra and taiga snowpack as well as a pan-Arctic synthesis that shows the decreasing sink strength of high-latitude terrestrial regions. Overall, our set of long-term, year-round measurements along this gradient of permafrost thaw are key in understanding changes in terrestrial carbon cycling in a vulnerable region.

List of Abbreviations

ABoVE	Arctic-Boreal Vulnerability Experiment, NASA
ACRC	Alaska Coastal Rainforest Center
ADF&G	Alaska Department of Fish and Game
AFS	American Fisheries Society
AFFD	Alaska Forage Fish Database
AKCFWRU	Alaska Cooperative Fish and Wildlife Research Unit, USGS
AKSG	Alaska Sea Grant, NOAA
BIA	U.S. Bureau of Indian Affairs
BLaST	Biomedical Learning and Student Training, UAF
BLM	U.S. Bureau of Land Management
BOEM	U.S. Bureau of Ocean Energy Management
CASC	Climate Adaptation Science Center, USGS
CESU	Cooperative Ecosystem Studies Unit, UAF
CFOS	College of Fisheries and Ocean Sciences, UAF
CRITFC	Columbia River Inter-Tribal Fish Commission
DBW	Department of Biology and Wildlife, UAF
DoD	U.S. Department of Defense
eDNA	environmental DNA
EPSCoR	Established Program to Stimulate Competitive Research, NSF
GPS	Global positioning system
GRFP	Graduate Research Fellowship Program, NSF
IAB	Institute of Arctic Biology, UAF
IALT	Interior Alaska Land Trust
IARC	International Arctic Research Center, UAF
IMS	Institute of Marine Science, UAF
INBRE	IDEA Networks of Biomedical Research Excellence, NIH
LTER	Long-Term Ecological Research Network, NSF
LTREB	Long Term Research in Environmental Biology, NSF
NASA	U.S. National Aeronautics and Space Administration
NCA	National Conservation Area
NIH	U.S. National Institutes of Health
NIWR	National Institutes for Water Resources
NOAA	U.S. National Oceanic and Atmospheric Administration
NPRB	U.S. North Pacific Research Board
NPS	U.S. National Park Service
NSF	National Science Foundation
NWR	National Wildlife Refuge
OSM	Office of Subsistence Management, USFWS
PI	Principal Investigator
PNWRS	Pacific Northwest Research Station, USFS
POW	Prince of Wales
R&M	Research and Monitoring program, USFWS

RSA	Reimbursable Services Agreement
RTS	Research and Technical Support, Division of Sport Fish, ADF&G
RWO	Research Work Order
SECR	spatially-explicit capture-recapture
SERDP	Strategic Environmental Research and Development Program, DoD
SNAP	Scenarios Network for Alaska + Arctic Planning
SSP	Science Support Program, USGS and USFWS
TCC	Tanana Chiefs Conference
TVWA	Tanana Valley Watershed Association
UAF	University of Alaska Fairbanks
UAS	University of Alaska Southeast
URSA	Undergraduate Research and Scholarly Activity program, UAF
USAG	U.S. Army Garrison
USDA	U.S. Department of Agriculture
USFS	U.S. Forest Service, USDA
USFWS	U.S. Fish and Wildlife Service
USGS	U.S. Geological Survey
VetMed	Veterinary Medicine Program, UAF
WERC	Water and Environmental Research Center, UAF